Mechanics III

Dynamics of Particles, Systems of Particles and Rigid Bodies

Lecture notes, written up from the handwritten notes of Lectures 1–13

Autumn Semester 2026

Preface

These notes cover the first thirteen lectures of Mechanics III. They follow the order and the numbering of the lectures (sections, examples), but whereas the handwritten notes taken in class are necessarily terse, the goal here is to tell the whole story: every result is motivated, every formula is derived, and every example is worked through to the end with its physical interpretation.

The course is built around a small number of balance laws. For a single particle, Newton's second law – the *balance of linear momentum* – is the only physical input; the *balance of angular momentum* and the *work–energy balance* are consequences of it, but each is the most convenient tool in a different kind of problem. The same three laws are then extended to systems of particles and finally to rigid bodies, where the angular momentum balance becomes genuinely independent and the moment of inertia tensor appears. Throughout, the most important practical skill is to choose the right balance law and the right reference point, so that unknown forces drop out.

Notation. Vectors are printed in bold, $\boldsymbol{r}$ (in handwriting and on the blackboard they are underlined, $\underline{r}$); second-order tensors are double underlined, $\underline{\underline{I}}$. For a vector $\boldsymbol{r}$, the column of its components in a frame C is written $[\boldsymbol{r}]_C$. A dot denotes the time derivative, $\dot{x} = \mathrm{d}x/\mathrm{d}t$. We use the abbreviations DOF (degree of freedom), FBD (free-body diagram), LMB, AMB and WEB for the balances of linear momentum, angular momentum and the work–energy balance, ICR for the instantaneous centre of rotation, IAR for the instantaneous axis of rotation, and cm for the centre of mass.

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Chapter 1

Dynamics of a Single Particle

Lecture 1
16.09.2026

Dynamics is the study of motion and its causes. We begin with the simplest possible model of a body, the particle, and develop the two halves of the theory one after the other. *Kinematics* (section 1.1) is the geometry of motion: how to describe positions, velocities and accelerations, without asking why the body moves the way it does. *Kinetics* (section 1.2) connects the motion to its cause, the forces, through the balance laws of mechanics.

Definition 1.1. A *particle* is an object of negligible size and shape. Mathematically it is a point carrying a mass m.

Whether this idealisation is appropriate depends on the question being asked, not on the size of the body: the earth is a particle when we compute its orbit around the sun, but certainly not when we discuss its rotation. What the definition rules out is precisely that: the orientation of a particle is meaningless, so only its position has to be described.

1.1 Kinematics

1.1.1 Cartesian reference frames

To describe a position we need a reference. Throughout the course the basic reference is a Cartesian frame.

Definition 1.2. A *Cartesian reference frame* C consists of a fixed origin O and a fixed orthonormal basis $\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3$, that is, $\boldsymbol{e}_i \cdot \boldsymbol{e}_j = \delta_{ij}$ and $\dot{\boldsymbol{e}}_i = \boldsymbol{0}$.

The word "fixed" is essential and will become important in section 1.1.3, where we meet bases that move. The position of the particle at time t is the vector from O to the particle,

$$\boldsymbol{r}(t) = \sum_{i=1}^{3} x_i(t)\, \boldsymbol{e}_i, \tag{1.1}$$

where the scalar functions $x_i(t)$ are its *coordinates*.

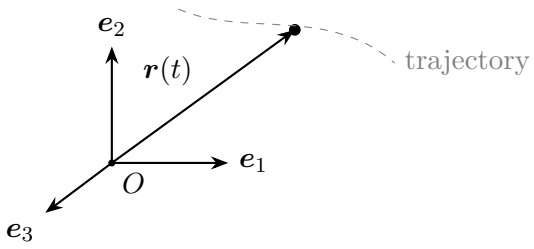


Figure 1.1: A particle in a Cartesian frame.

Each coordinate that can be chosen independently is called a *degree of freedom* (DOF). For a particle that is free to move, all its coordinates are independent; and since the orientation of a particle does not matter, there are no further DOFs.

Proposition 1.3. *A free particle in d dimensions has d degrees of freedom.*

Velocity and acceleration. The velocity is the rate of change of the position, and the acceleration the rate of change of the velocity. Because the basis vectors of a Cartesian frame are constant, the time derivative acts only on the coordinates:

$$\boldsymbol{v} = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} = \sum_{i=1}^{d} \dot{x}_i \, \boldsymbol{e}_i, \qquad \boldsymbol{a} = \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = \sum_{i=1}^{d} \ddot{x}_i \, \boldsymbol{e}_i. \tag{1.2}$$

It is worth pausing on the geometric content of these definitions. During an infinitesimal time step $\mathrm{d}t$ the particle moves by $\mathrm{d}\boldsymbol{r} = \boldsymbol{v}\,\mathrm{d}t$ and its velocity changes by $\mathrm{d}\boldsymbol{v} = \boldsymbol{a}\,\mathrm{d}t$. The first relation says that the velocity points in the direction of $\mathrm{d}\boldsymbol{r}$: it is *tangent to the trajectory*, pointing where the particle is going. The second relation says that the acceleration points in the direction of the *change* of the velocity. In general this has nothing to do with the direction of motion: a car driving through a bend at constant speed accelerates sideways (fig. 1.2).

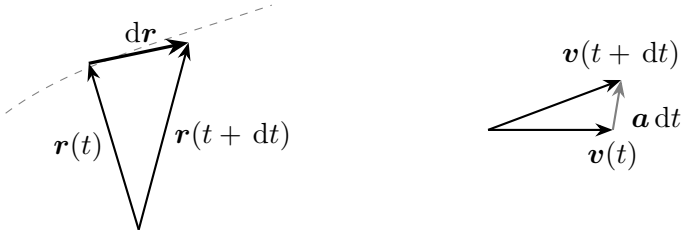


Figure 1.2: The velocity is tangent to the trajectory; the acceleration points in the direction of the change of velocity.

The *speed* is the magnitude of the velocity, $v = |\boldsymbol{v}| = \sqrt{\boldsymbol{v}\cdot\boldsymbol{v}}$. Collecting coordinates in columns we obtain the component form

$$[\boldsymbol{r}]_C = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \qquad [\boldsymbol{v}]_C = \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{pmatrix}, \qquad [\boldsymbol{a}]_C = \begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{pmatrix}. \tag{1.3}$$

The subscript is a reminder that a vector is a geometric object independent of any frame, whereas its components are not: the same vector has different columns in different frames.

From acceleration to trajectory. In kinetics, the balance laws will typically tell us the acceleration; the task is then to reconstruct the motion. The following examples show how this is done in the most common situations: the acceleration is known as a function of time, of velocity, or of position.

Example 1.1 *(Finding $\boldsymbol{r}(t)$ from a known $\boldsymbol{a}(t)$).* Starting from $\boldsymbol{a} = \mathrm{d}\boldsymbol{v}/\mathrm{d}t$ we separate the differentials and integrate from the initial instant t_0 to the current time t:

$$\int_{t_0}^{t} \boldsymbol{a}(\tau)\,\mathrm{d}\tau = \int_{\boldsymbol{v}(t_0)}^{\boldsymbol{v}(t)} \mathrm{d}\boldsymbol{v} \qquad \Longrightarrow \qquad \boldsymbol{v}(t) = \boldsymbol{v}(t_0) + \int_{t_0}^{t} \boldsymbol{a}(\tau)\,\mathrm{d}\tau.$$

Using definite integrals with the initial state as the lower limit has the advantage that no integration constant has to be determined afterwards. In exactly the same way, $\boldsymbol{v} = \mathrm{d}\boldsymbol{r}/\mathrm{d}t$ gives

$$\boldsymbol{r}(t) = \boldsymbol{r}(t_0) + \int_{t_0}^{t} \boldsymbol{v}(\tau)\,\mathrm{d}\tau.$$

As an application, consider free fall. With $\boldsymbol{e}_2$ pointing upwards, the acceleration due to gravity is $\boldsymbol{a} = -g\boldsymbol{e}_2 = \text{const}$. Taking $t_0 = 0$,

$$\boldsymbol{v}(t) = \boldsymbol{v}(0) - gt\,\boldsymbol{e}_2, \qquad \boldsymbol{r}(t) = \boldsymbol{r}(0) + \boldsymbol{v}(0)\,t - \frac{gt^2}{2}\,\boldsymbol{e}_2,$$

or in components

$$\begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{pmatrix} = \begin{pmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \end{pmatrix} + \begin{pmatrix} \dot{x}_1(0) \\ \dot{x}_2(0) \\ \dot{x}_3(0) \end{pmatrix} t + \begin{pmatrix} 0 \\ -gt^2/2 \\ 0 \end{pmatrix}.$$

Since the equation of motion is of second order, the motion is determined only once *two* initial conditions are given, the initial position $\boldsymbol{r}(0)$ and the initial velocity $\boldsymbol{v}(0)$. ◀

Example *(Clicker question: which particle lands first?).* Two identical particles are launched at the same time. Particle 1 starts from the ground vertically upwards with speed v_0; particle 2 is released from rest at height h. Both move with the constant acceleration g downwards. From what height must particle 2 be released for particle 1 to land first?

By the previous example, the height of particle 1 is $x_2^{(1)}(t) = v_0 t - \frac{1}{2}gt^2 = t(v_0 - \frac{1}{2}gt)$. It vanishes at $t = 0$ (the launch) and at $t^{(1)} = 2v_0/g$ (the landing). Particle 2 has $x_2^{(2)}(t) = h - \frac{1}{2}gt^2$, which vanishes at $t^{(2)} = \sqrt{2h/g}$. Particle 1 lands first if $t^{(1)} < t^{(2)}$; both sides are positive, so we may square:

$$\frac{4v_0^2}{g^2} < \frac{2h}{g} \quad\Longleftrightarrow\quad h > \frac{2v_0^2}{g}.$$

As a plausibility check, particle 1 rises to $v_0^2/(2g)$, so particle 2 has to start four times as high. ◀

Example 1.5 *(Finding the motion from $a(v)$ or $a(x)$). (1) Drag.* A particle moving through a medium is slowed down by it. For slow motion the drag is well described by Stokes' law, which makes the acceleration proportional to the velocity, $\boldsymbol{a} = -k\boldsymbol{v}$ with $k > 0$. In one dimension ($\boldsymbol{r} = x\boldsymbol{e}_1$, $\boldsymbol{v} = v\boldsymbol{e}_1$, $\boldsymbol{a} = a\boldsymbol{e}_1$) we have $a = a(v) = -kv$. The time does not appear explicitly, so we separate variables in $a(v) = \mathrm{d}v/\mathrm{d}t$:

$$\int_{t_0}^{t} \mathrm{d}\tau = \int_{v(t_0)}^{v(t)} \frac{\mathrm{d}v}{a(v)} = -\frac{1}{k}\int_{v(t_0)}^{v(t)} \frac{\mathrm{d}v}{v} \qquad\Longrightarrow\qquad t - t_0 = -\frac{1}{k}\ln\frac{v(t)}{v(t_0)},$$

that is, $v(t) = v(t_0)\,e^{-k(t-t_0)}$. The velocity decays exponentially; one more integration gives $x(t) = x(t_0) + \frac{v(t_0)}{k}\bigl(1 - e^{-k(t-t_0)}\bigr)$, so the particle comes to rest after the finite distance $v(t_0)/k$, although it needs infinitely long to do so.

(2) Rocket launch. Far from the surface of the earth, the gravitational acceleration depends on the altitude: $a = a(x)$. Now the trick is to eliminate the time by the chain rule,

$$a(x) = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}v}{\mathrm{d}x}\,\frac{\mathrm{d}x}{\mathrm{d}t} = v\,\frac{\mathrm{d}v}{\mathrm{d}x},$$

which separates into $v\,\mathrm{d}v = a(x)\,\mathrm{d}x$ and integrates to

$$\tfrac{1}{2}(v(x)^2 - v(x_0)^2) = \int_{x_0}^{x} a(\xi)\,\mathrm{d}\xi.$$

This gives the velocity as a function of position, $v = v(x)$. To obtain the position as a function of time, we separate once more: $\mathrm{d}t = \mathrm{d}x/v(x)$, hence $t - t_0 = \int_{x_0}^{x} \mathrm{d}\xi/v(\xi)$, which can be inverted to give $x(t)$. The relation $\frac{1}{2}\mathrm{d}(v^2) = a\,\mathrm{d}x$ obtained here is a first glimpse of the work–energy balance of section 1.2.3. ◀

1.1.2 Constrained motion

Very often a particle is not free: a bead on a wire, a block on a table, a mass on a string. Each such condition is a *kinematic constraint*, a relation that the coordinates must satisfy at all times. Each independent constraint allows us to express one coordinate in terms of the others and therefore removes one degree of freedom.

Proposition 1.4. *A particle in d dimensions subject to k independent kinematic constraints has $d - k$ degrees of freedom.*

Example 1.6 *((Un-)constrained motion).* A particle moving freely in space, $\boldsymbol{r} = \sum_{i=1}^{d} x_i \boldsymbol{e}_i$, has d DOFs. A particle on a rigid plane with normal $\boldsymbol{n}$ must satisfy $\boldsymbol{r} \cdot \boldsymbol{n} = \text{const}$; in three dimensions it therefore has $3 - 1 = 2$ DOFs, and choosing $\boldsymbol{e}_1, \boldsymbol{e}_2$ in the plane we can write $\boldsymbol{r} = x_1\boldsymbol{e}_1 + x_2\boldsymbol{e}_2$. A particle attached to a fixed point by a rigid link of length L satisfies $|\boldsymbol{r}| = L$ and has $d - 1$ DOFs; it moves on a circle in two dimensions and on a sphere in three. Finally, a particle confined to a tube or a track has a single DOF: its position is completely described by the path length s travelled along the track. ◀

Constraints do not enforce themselves. Whatever keeps the particle on the plane, on the circle or in the tube is a force, and so we note the following principle, which will be used in every problem of kinetics: *every constraint gives rise to a reaction force* – the normal force of a surface, the tension in a string – which is exactly the force needed to enforce the constraint. Conversely, a constraint that changes the direction of the velocity produces an acceleration (even at constant speed), and by Newton's law this requires a force.

1.1.3 Kinematics in polar coordinates

LECTURE 2
21.09.2026

Many motions are naturally described by a distance and an angle: a mass on a string, a planet around the sun, a bead on a rotating rod. In two dimensions we therefore describe the position by the distance $r(t)$ from the origin and the angle $\varphi(t)$ measured from $\boldsymbol{e}_1$:

$$\boldsymbol{r}(t) = r\cos\varphi\,\boldsymbol{e}_1 + r\sin\varphi\,\boldsymbol{e}_2. \tag{1.4}$$

We could differentiate this directly. The product and chain rules give

$$\boldsymbol{v} = (\dot{r}\cos\varphi - r\dot{\varphi}\sin\varphi)\boldsymbol{e}_1 + (\dot{r}\sin\varphi + r\dot{\varphi}\cos\varphi)\boldsymbol{e}_2,$$

and differentiating once more produces a rather unpleasant expression for the acceleration, whose physical meaning is completely hidden. The remedy is to use a basis adapted to the motion.

Definition 1.5. The *polar basis* associated with the angle φ is

$$\boldsymbol{e}_r(t) = \cos\varphi\,\boldsymbol{e}_1 + \sin\varphi\,\boldsymbol{e}_2, \qquad \boldsymbol{e}_\varphi(t) = -\sin\varphi\,\boldsymbol{e}_1 + \cos\varphi\,\boldsymbol{e}_2. \tag{1.5}$$

The vector $\boldsymbol{e}_r$ points radially outwards towards the particle, and $\boldsymbol{e}_\varphi$ is obtained from it by a quarter turn in the direction of increasing φ. One checks immediately that $|\boldsymbol{e}_r| = |\boldsymbol{e}_\varphi| = 1$ and $\boldsymbol{e}_r \cdot \boldsymbol{e}_\varphi = 0$, so the polar basis is orthonormal. The crucial difference to a Cartesian basis is that it co-rotates with the particle and is therefore *time dependent*.

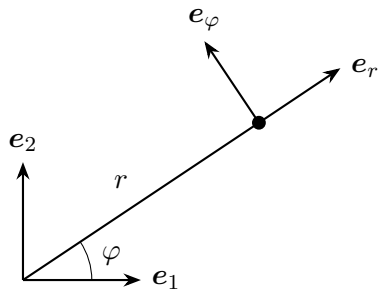


Figure 1.3: The co-rotating polar basis.

Since the basis moves, we need its time derivatives. Differentiating with the chain rule,

$$\dot{\boldsymbol{e}}_r = -\sin\varphi\,\dot\varphi\,\boldsymbol{e}_1 + \cos\varphi\,\dot\varphi\,\boldsymbol{e}_2 = \dot\varphi\,\boldsymbol{e}_\varphi, \tag{1.6}$$

$$\dot{\boldsymbol{e}}_\varphi = -\cos\varphi\,\dot\varphi\,\boldsymbol{e}_1 - \sin\varphi\,\dot\varphi\,\boldsymbol{e}_2 = -\dot\varphi\,\boldsymbol{e}_r. \tag{1.7}$$

These two identities are all we need. They are easy to remember geometrically: a unit vector cannot change its length, so its derivative is perpendicular to it, and its magnitude is the rate $\dot\varphi$ at which the vector turns.

With the polar basis the position takes the compact form $\boldsymbol{r} = r\,\boldsymbol{e}_r$. Differentiating with the product rule and using (1.6),

$$\boldsymbol{v} = \dot r\,\boldsymbol{e}_r + r\,\dot{\boldsymbol{e}}_r = \dot r\,\boldsymbol{e}_r + r\dot\varphi\,\boldsymbol{e}_\varphi. \tag{1.8}$$

The two terms have a clear meaning: the radial component $\dot r$ comes from the change of distance, the circumferential component $r\dot\varphi$ from the rotation. We call $\omega \coloneqq \dot\varphi$ the *angular velocity* (unit $\mathrm{rad\,s^{-1}}$) and $\dot\omega = \ddot\varphi$ the *angular acceleration* ($\mathrm{rad\,s^{-2}}$). Differentiating (1.8) once more, now using both (1.6) and (1.7),

$$\begin{aligned}\boldsymbol{a} &= \ddot r\,\boldsymbol{e}_r + \dot r\,\dot{\boldsymbol{e}}_r + \dot r\dot\varphi\,\boldsymbol{e}_\varphi + r\ddot\varphi\,\boldsymbol{e}_\varphi + r\dot\varphi\,\dot{\boldsymbol{e}}_\varphi \\ &= \ddot r\,\boldsymbol{e}_r + \dot r\dot\varphi\,\boldsymbol{e}_\varphi + \dot r\dot\varphi\,\boldsymbol{e}_\varphi + r\ddot\varphi\,\boldsymbol{e}_\varphi - r\dot\varphi^2\,\boldsymbol{e}_r,\end{aligned}$$

and collecting terms we arrive at

$$\boldsymbol{a} = \big(\ddot r - r\dot\varphi^2\big)\boldsymbol{e}_r + \big(2\dot r\dot\varphi + r\ddot\varphi\big)\boldsymbol{e}_\varphi. \tag{1.9}$$

Each term has a name and a meaning. The term $-r\dot\varphi^2$ is the *centripetal acceleration*; it always points towards the origin and is present whenever the particle rotates, even at constant r and $\dot\varphi$. The term $2\dot r\dot\varphi$ is the *Coriolis acceleration*: if you walk outwards on a rotating platform, your circumferential speed $r\dot\varphi$ grows with r, so you must be accelerated sideways. Finally, $r\ddot\varphi$ is the tangential acceleration caused by a change of the rotation rate.

Remark 1.6. Note that $v_r = \dot r$, but $a_r \neq \dot v_r$, and likewise for the φ-components. Because the basis is time dependent, the components of the acceleration are *not* the derivatives of the components of the velocity. Forgetting this is one of the most common mistakes in polar coordinates.

Example 1.7 *(Particle on a taut, inextensible string).* The particle moves on a circle of radius L, so $r = L = \text{const}$ and $\dot r = \ddot r = 0$. Equations (1.8) and (1.9) reduce to

$$\boldsymbol{v} = L\dot\varphi\,\boldsymbol{e}_\varphi, \qquad \boldsymbol{a} = -L\dot\varphi^2\,\boldsymbol{e}_r + L\ddot\varphi\,\boldsymbol{e}_\varphi,$$

and the speed is $|\boldsymbol{v}| = L|\omega|$. For uniform rotation, $\omega = \text{const}$, only the centripetal term survives: $\boldsymbol{a} = -L\omega^2\boldsymbol{e}_r = -(v^2/L)\boldsymbol{e}_r$. The particle accelerates although its speed is constant, because the direction of its velocity keeps changing. ◀

Example *(Clicker question: moving radially on a rotating disk).* A disk rotates with constant angular velocity ω, and a particle moves radially inwards on it with constant speed v_0; thus $\dot r = -v_0$, $\ddot r = 0$, $\dot\varphi = \omega$, $\ddot\varphi = 0$. Then

$$\boldsymbol{v} = -v_0\boldsymbol{e}_r + r\omega\,\boldsymbol{e}_\varphi, \qquad \boldsymbol{a} = -r\omega^2\boldsymbol{e}_r - 2v_0\omega\,\boldsymbol{e}_\varphi, \qquad |\boldsymbol{a}| = \sqrt{(r\omega^2)^2 + (2v_0\omega)^2}.$$

Since v_0 and ω are constant, the magnitude of the acceleration is smallest at the centre $r = 0$, where it equals $2|v_0\omega|$. The Coriolis part never vanishes. ◀

1.1.4 Kinematics of space curves

When the path of the particle is prescribed – a road, a rail, a roller coaster track – the most natural coordinate is the *path length* $s(t)$ measured along the curve from some reference point, and the most natural basis is attached to the curve itself. The *unit tangent* $\boldsymbol{e}_t = \boldsymbol{v}/|\boldsymbol{v}|$ points in the direction of motion, so that

$$\boldsymbol{v} = v\,\boldsymbol{e}_t = \dot s\,\boldsymbol{e}_t, \tag{1.10}$$

where we have used that the speed is the rate at which path length is covered, $v = |\boldsymbol{v}| = \dot s$.

The acceleration follows from the product rule, $\boldsymbol{a} = \ddot s\,\boldsymbol{e}_t + \dot s\,\dot{\boldsymbol{e}}_t$, and the question is how fast the tangent turns. Locally, every smooth curve can be approximated by a circle, the *osculating circle*, whose radius ρ is the *radius of curvature*. Along a circle of radius ρ a path element $\mathrm{d}s$ corresponds to a change of direction $\mathrm{d}\theta = \mathrm{d}s/\rho$, so the tangent changes at the rate $\mathrm{d}\boldsymbol{e}_t/\mathrm{d}s = \boldsymbol{e}_n/\rho$, where the *unit normal* $\boldsymbol{e}_n$ points towards the centre of the osculating circle. Hence $\dot{\boldsymbol{e}}_t = (\dot s/\rho)\boldsymbol{e}_n$ and

$$\boldsymbol{a} = \ddot s\,\boldsymbol{e}_t + \frac{v^2}{\rho}\,\boldsymbol{e}_n. \tag{1.11}$$

The first term changes the speed along the path; the second, always directed towards the centre of curvature, changes the direction. On a straight piece of track $\rho \to \infty$, the normal term disappears and $\boldsymbol{a} = \ddot s\,\boldsymbol{e}_t$.

Example 1.10 *(Circular motion).* On a circle of radius R we have $\rho = R$ and $s = R\varphi$, so (1.11) gives $\boldsymbol{v} = R\dot\varphi\,\boldsymbol{e}_t$ and $\boldsymbol{a} = R\ddot\varphi\,\boldsymbol{e}_t + (v^2/R)\,\boldsymbol{e}_n$. Polar coordinates with $r = R$ gave $\boldsymbol{a} = R\ddot\varphi\,\boldsymbol{e}_\varphi - (v^2/R)\,\boldsymbol{e}_r$. The two results agree: $\boldsymbol{e}_t = \boldsymbol{e}_\varphi$, and since $\boldsymbol{e}_n$ points towards the centre while $\boldsymbol{e}_r$ points away from it, $\boldsymbol{e}_n = -\boldsymbol{e}_r$. ◀

Formula (1.11) has an immediate practical consequence for track design. If a straight track is joined directly to a circular arc of radius R, the curvature jumps from 0 to $1/R$ at the junction, and so does the normal acceleration v^2/ρ – passengers are hit by a sudden sideways jolt. Good designs insert a transition curve along which the radius of curvature decreases gradually (for instance a clothoid, whose curvature grows linearly with s), so that v^2/ρ builds up smoothly; in a looping one aims to keep v^2/ρ roughly constant, which is why real loopings are not circular.

1.2 Kinetics

LECTURE 3
23.09.2026

We now turn to the causes of motion. The central quantity is the momentum of the particle.

Definition 1.7. The *linear momentum* of a particle of mass m and velocity $\boldsymbol{v}$ is $\boldsymbol{p} = m\boldsymbol{v}$.

1.2.1 Balance of linear momentum

The fundamental law of mechanics, Newton's second law, states how the momentum changes.

Axiom 1.8 (Balance of linear momentum)**.** *In an inertial frame, the rate of change of the momentum of a particle equals the net force acting on it:*

$$\sum_i \boldsymbol{F}_i = \frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(m\boldsymbol{v}). \tag{1.12}$$

Several familiar statements are special cases. In *statics* nothing moves, $\boldsymbol{v} = \boldsymbol{0}$, so $\boldsymbol{p} = \boldsymbol{0}$ and the forces must balance, $\sum_i \boldsymbol{F}_i = \boldsymbol{0}$. Conversely, if the net force vanishes, the momentum is constant and so is the velocity of a particle of constant mass: it moves uniformly along a straight line. This is *Newton's first axiom.* If the mass is constant, $\dot{m} = 0$, then $\dot{\boldsymbol{p}} = m\dot{\boldsymbol{v}}$ and we obtain *Newton's second axiom* in its usual form,

$$\sum_i \boldsymbol{F}_i = m\boldsymbol{a}. \tag{1.13}$$

It is the acceleration, not the velocity, that is determined by the forces. Finally, for a massless object ($m = 0$) the momentum is always zero, so the forces on it must balance *even when it accelerates.* This is why we may treat light ropes, pulleys and links as if they were in static equilibrium.

The third axiom completes the picture.

Axiom 1.9 (Newton's third axiom, "actio = reactio")**.** *If a body A exerts a force $\boldsymbol{F}$ on a body B, then B exerts the force $-\boldsymbol{F}$ on A.*

This axiom is what makes free-body diagrams consistent. To apply (1.12) to one body of a mechanism, we cut it free from its surroundings and replace every contact by the corresponding force; the same force then appears with the opposite sign on the body on the other side of the cut.

Example *(Free-body diagram of a mass on a block).* A small mass m lies on a block of mass M, which rests on the ground; $\boldsymbol{e}_2$ points up. Cutting the mass free, it

experiences its weight $-mg\boldsymbol{e}_2$ and the contact force $N_1\boldsymbol{e}_2$ from the block. The block experiences its weight $-Mg\boldsymbol{e}_2$, the reaction $-N_1\boldsymbol{e}_2$ of the mass, and the force $N_2\boldsymbol{e}_2$ from the ground. (The ground actually acts through a distributed pressure, but only its resultant enters the momentum balance.) Applying (1.12) in the vertical direction to each body,

$$N_1 - mg = ma_1, \qquad N_2 - N_1 - Mg = Ma_2.$$

◀

Example 1.12 *(Block sliding on frictionless ground).* The block can only move horizontally, so $\boldsymbol{r} = x_1\boldsymbol{e}_1$, $\boldsymbol{v} = \dot{x}_1\boldsymbol{e}_1$, $\boldsymbol{a} = \ddot{x}_1\boldsymbol{e}_1$. The friction coefficient is $\mu = 0$, so the only forces are gravity and the normal force, and the momentum balance reads

$$(N - mg)\,\boldsymbol{e}_2 = m\ddot{x}_1\,\boldsymbol{e}_1.$$

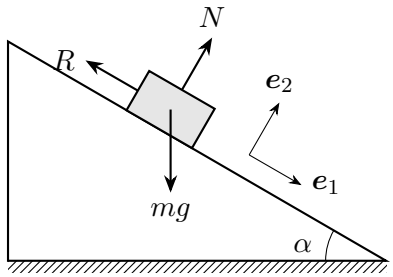
The left-hand side has only an $\boldsymbol{e}_2$-component and the right-hand side only an $\boldsymbol{e}_1$-component, so both must vanish: $N = mg$ and $\ddot{x}_1 = 0$. The block slides on with constant velocity. ◀

Real surfaces resist sliding. In the Coulomb model of friction, as long as the surfaces stick, the friction force $\boldsymbol{R}$ adjusts itself to whatever value is needed to prevent sliding, but only up to a limit, $|\boldsymbol{R}| \le \mu_0|\boldsymbol{N}|$, where μ_0 is the static friction coefficient. Once the surfaces slide, the friction force has a fixed magnitude and opposes the sliding velocity:

$$\boldsymbol{R} = -\mu|\boldsymbol{N}|\,\frac{\boldsymbol{v}}{|\boldsymbol{v}|}, \tag{1.14}$$

with the dynamic friction coefficient μ.

Example 1.13 *(Sliding down a rough incline).*



We choose $\boldsymbol{e}_1$ down the slope and $\boldsymbol{e}_2$ normal to it, so that the kinematics is again one-dimensional, $\boldsymbol{a} = \ddot{x}_1\boldsymbol{e}_1$. The block slides downwards, so the dynamic friction force points up the slope, $\boldsymbol{R} = -R\boldsymbol{e}_1$ with $R = \mu N$. Decomposing the weight into the components $mg\sin\alpha$ along $\boldsymbol{e}_1$ and $-mg\cos\alpha$ along $\boldsymbol{e}_2$, the momentum balance is

$$(mg\sin\alpha - R)\,\boldsymbol{e}_1 + (N - mg\cos\alpha)\,\boldsymbol{e}_2 = m\ddot{x}_1\,\boldsymbol{e}_1.$$

The $\boldsymbol{e}_2$-component gives the normal force, $N = mg\cos\alpha$, and the $\boldsymbol{e}_1$-component together with $R = \mu N$ gives the acceleration

$$\ddot{x}_1 = g\big(\sin\alpha - \mu\cos\alpha\big).$$

If $\tan\alpha < \mu$ this is negative: the block decelerates and eventually comes to rest. ◀

Example 1.14 *(Projectile motion).* If gravity is the only force, the momentum balance $m\boldsymbol{a} = m\boldsymbol{g}$ gives $\boldsymbol{a} = \boldsymbol{g}$, independently of the mass. The horizontal velocity therefore stays exactly the same during the whole flight, while the vertical motion is the free fall of Example 1.1; the trajectory is a parabola. ◀

Example 1.15 *(Particle on a string).* A mass m on a string of constant length L rotates about a fixed point with constant angular velocity ω. From Example 1.7, $\boldsymbol{a} = -L\omega^2\boldsymbol{e}_r$. The only force (we ignore gravity) is the string force, which pulls towards the centre, $-S\boldsymbol{e}_r$. The momentum balance $-S\boldsymbol{e}_r = -mL\omega^2\boldsymbol{e}_r$ thus gives $S = mL\omega^2$: the string has to supply exactly the centripetal force. ◀

1.2.2 Balance of angular momentum

For rotational motion it is useful to consider the moment of the momentum about a point.

Definition 1.10. The *angular momentum* of a particle P with respect to a reference point B is

$$\boldsymbol{H}_B = \boldsymbol{r}_{BP} \times \boldsymbol{p}, \qquad \boldsymbol{r}_{BP} = \boldsymbol{r}_P - \boldsymbol{r}_B, \quad \boldsymbol{p} = m\boldsymbol{v}_P. \tag{1.15}$$

Theorem 1.11 (Balance of angular momentum)**.** *Let $\boldsymbol{M}_B = \boldsymbol{r}_{BP} \times \sum_i \boldsymbol{F}_i$ be the net torque about B. Then*

$$\boldsymbol{M}_B = \dot{\boldsymbol{H}}_B + \boldsymbol{v}_B \times \boldsymbol{p}. \tag{1.16}$$

Derivation. By the product rule, $\dot{\boldsymbol{H}}_B = \dot{\boldsymbol{r}}_{BP} \times \boldsymbol{p} + \boldsymbol{r}_{BP} \times \dot{\boldsymbol{p}}$. In the first term $\dot{\boldsymbol{r}}_{BP} = \boldsymbol{v}_P - \boldsymbol{v}_B$, and $\boldsymbol{v}_P \times \boldsymbol{p} = m\,\boldsymbol{v}_P \times \boldsymbol{v}_P = \boldsymbol{0}$, so it equals $-\boldsymbol{v}_B \times \boldsymbol{p}$. In the second term we use the momentum balance $\dot{\boldsymbol{p}} = \sum_i \boldsymbol{F}_i$, which turns it into the torque $\boldsymbol{M}_B$. Hence $\dot{\boldsymbol{H}}_B = -\boldsymbol{v}_B \times \boldsymbol{p} + \boldsymbol{M}_B$. □

For a single particle the angular momentum balance contains no new physics – we have just derived it from the momentum balance. Its value lies in its convenience: forces whose line of action passes through B have no torque about B and simply drop out, so a clever choice of B eliminates unknown reaction forces.

The extra term $\boldsymbol{v}_B \times \boldsymbol{p}$ disappears if the reference point is fixed, $\boldsymbol{v}_B = \boldsymbol{0}$, or if it moves parallel to the particle. In that case the balance takes the simple form $\boldsymbol{M}_B = \dot{\boldsymbol{H}}_B$, and the reference point should be chosen in this way whenever possible. In particular, if there is no torque about such a point, the angular momentum about it is conserved. In statics, and for a massless particle, both $\boldsymbol{H}_B$ and $\boldsymbol{p}$ vanish and the balance reduces to $\boldsymbol{M}_B = \boldsymbol{0}$.

Plane motion about a fixed point. In two dimensions, with B fixed at the origin of polar coordinates, $\boldsymbol{r}_{BP} = r\boldsymbol{e}_r$ and $\boldsymbol{v}_P = v_r\boldsymbol{e}_r + v_\varphi\boldsymbol{e}_\varphi$. Since $\boldsymbol{e}_r \times \boldsymbol{e}_r = \boldsymbol{0}$ and $\boldsymbol{e}_r \times \boldsymbol{e}_\varphi = \boldsymbol{e}_3$,

$$\boldsymbol{H}_B = m\,r\boldsymbol{e}_r \times (v_r\boldsymbol{e}_r + v_\varphi\boldsymbol{e}_\varphi) = mrv_\varphi\,\boldsymbol{e}_3 = mr^2\omega\,\boldsymbol{e}_3.$$

The angular momentum always points out of the plane, and writing $\boldsymbol{M}_B = M_B\boldsymbol{e}_3$ the balance becomes a scalar equation,

$$M_B = \frac{\mathrm{d}}{\mathrm{d}t}(mr^2\omega). \tag{1.17}$$

If, moreover, the distance is constant, $r = R$, this reduces to

$$M_B = mR^2\dot{\omega} = I_B\ddot{\varphi}, \qquad I_B = mR^2. \tag{1.18}$$

The quantity I_B is the *mass moment of inertia* about B. It plays the same role for rotation that the mass plays for translation – compare $F = ma$ with $M_B = I_B\ddot{\varphi}$ – and measures the resistance of the body against angular acceleration. We shall meet it again, in much greater generality, in chapter 3.

1.2.3 Work–energy balance

LECTURE 4
28.09.2026

The momentum balance is a differential equation in time. In many problems, however, we are not interested in *when* something happens but only in the relation between the state of motion at two different positions – how fast is the particle at the bottom of the hill? For such questions an integrated form of the momentum balance is far more efficient.

Theorem 1.12 (Work–energy balance)**.** *For a particle of constant mass moving from $\boldsymbol{r}_1 = \boldsymbol{r}(t_1)$ to $\boldsymbol{r}_2 = \boldsymbol{r}(t_2)$,*

$$T(t_2) - T(t_1) = W_{12}, \qquad T = \frac{m}{2}|\boldsymbol{v}|^2, \qquad W_{12} = \int_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} \sum_i \boldsymbol{F}_i \cdot \mathrm{d}\boldsymbol{r}. \tag{1.19}$$

T is the kinetic energy *and W_{12} the* work *done by the forces on the particle: the change of kinetic energy equals the work done.*

Derivation. Take the scalar product of the momentum balance with $\mathrm{d}\boldsymbol{r}$ and integrate along the path:

$$\int_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} \frac{\mathrm{d}}{\mathrm{d}t}(m\boldsymbol{v}) \cdot \mathrm{d}\boldsymbol{r} = \int_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} \sum_i \boldsymbol{F}_i \cdot \mathrm{d}\boldsymbol{r} = W_{12}.$$

On the left we substitute $\mathrm{d}\boldsymbol{r} = \boldsymbol{v}\,\mathrm{d}t$ and change the integration variable:

$$m\int_{t_1}^{t_2} \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} \cdot \boldsymbol{v}\,\mathrm{d}t = m\int_{\boldsymbol{v}(t_1)}^{\boldsymbol{v}(t_2)} \boldsymbol{v} \cdot \mathrm{d}\boldsymbol{v} = \frac{m}{2}|\boldsymbol{v}(t_2)|^2 - \frac{m}{2}|\boldsymbol{v}(t_1)|^2. \qquad \square$$

An important observation is that a force perpendicular to the motion, $\boldsymbol{F} \perp \mathrm{d}\boldsymbol{r}$, does no work. The normal force of a fixed surface and the string force of a pendulum are of this kind, and so they disappear from the work–energy balance altogether. This is what makes it so effective.

Example 1.21 *(Frictional sliding).* A block with initial velocity v_0 slides on rough ground. Where does it come to rest? The kinetic energy is $\frac{1}{2}mv_0^2$ initially and zero at the end. Of the forces acting on the block, gravity and the normal force are perpendicular to the motion $\mathrm{d}\boldsymbol{r} = \mathrm{d}x_1\boldsymbol{e}_1$ and do no work; only friction, $-\mu N\boldsymbol{e}_1 = -\mu mg\boldsymbol{e}_1$, does. Thus $W_{12} = -\mu mg\,\Delta x$, and (1.19) gives

$$0 - \tfrac{1}{2}mv_0^2 = -\mu mg\,\Delta x \qquad \Longrightarrow \qquad \Delta x = \frac{v_0^2}{2\mu g}.$$

We never needed to know how long the block slides. ◀

Conservative forces. Computing the work requires, in general, knowledge of the path. For an important class of forces it does not.

Definition 1.13. A force $\boldsymbol{F}$ is *conservative* if there is a function $V(\boldsymbol{r})$, the *potential energy*, such that $\boldsymbol{F} = -\mathrm{d}V/\mathrm{d}\boldsymbol{r} = -\nabla V$.

For instance, gravity is conservative: with $\boldsymbol{e}_2$ up, $V(\boldsymbol{r}) = mg\,\boldsymbol{r}\cdot\boldsymbol{e}_2 = mgx_2$ gives $\boldsymbol{F} = -\nabla V = -mg\boldsymbol{e}_2$.

Proposition 1.14. *The work of a conservative force depends only on the end points of the path: $W_{12} = V(\boldsymbol{r}_1) - V(\boldsymbol{r}_2)$.*

Proof. $W_{12} = \int_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} \boldsymbol{F}\cdot\mathrm{d}\boldsymbol{r} = -\int_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} \nabla V\cdot\mathrm{d}\boldsymbol{r} = -\int_{V(\boldsymbol{r}_1)}^{V(\boldsymbol{r}_2)} \mathrm{d}V$. □

We can check this on a particle sliding down a frictionless curved track from height h_1 to h_2. The normal force is perpendicular to the velocity, so only gravity does work, $W_{12} = \int(-mg\boldsymbol{e}_2)\cdot\mathrm{d}\boldsymbol{r} = -mg\int\mathrm{d}x_2 = -mg(h_2-h_1) = V_1 - V_2$, independently of the shape of the track.

If we now split the forces into conservative forces, which we account for by their potential, and the remaining non-conservative ones, the work is $W_{12} = V(\boldsymbol{r}_1) - V(\boldsymbol{r}_2) + W_{12}^{\mathrm{nc}}$, and the work–energy balance takes the form

$$T(t_2) + V(\boldsymbol{r}_2) - \big[T(t_1) + V(\boldsymbol{r}_1)\big] = W_{12}^{\mathrm{nc}}, \qquad W_{12}^{\mathrm{nc}} = \int_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} \sum_i \boldsymbol{F}_i^{\mathrm{nc}}\cdot\mathrm{d}\boldsymbol{r}. \tag{1.20}$$

A system in which work is done only by conservative forces is called *conservative*. For it $W_{12}^{\mathrm{nc}} = 0$, and the total mechanical energy is conserved:

$$T + V = \text{const.} \tag{1.21}$$

Example 1.22 *(Maximum flight height).* A projectile is launched with speed v_0 at an angle α to the horizontal; there is no air drag. At the highest point the vertical velocity vanishes, while the horizontal velocity $v_0\cos\alpha$ is unchanged (Example 1.14). Conservation of energy between launch and top gives

$$\frac{m}{2}v_0^2 = \frac{m}{2}(v_0\cos\alpha)^2 + mgx_{2,\max} \qquad\Longrightarrow\qquad x_{2,\max} = \frac{v_0^2}{2g}(1-\cos^2\alpha) = \frac{v_0^2}{2g}\sin^2\alpha.$$

◀

Example 1.23 *(Car in a looping).* A car, modelled as a particle of mass m, starts from rest at height h_0 and rolls without friction and without engine power down a ramp into a circular looping of radius R. We ask for the minimal height h_0 for which the car completes the looping, and for the largest force it experiences.

Energy conservation gives the speed at height h: $mgh_0 = mgh + \tfrac12 mv^2$, so $v^2 = 2g(h_0 - h)$. Requiring $v^2 \geq 0$ at the top of the looping yields $h_0 \geq 2R$. This condition is necessary, but not sufficient: with h_0 only slightly above $2R$, the car would still have speed left at the top, but long before reaching it it would lose contact with the track and fall. The real condition is that the track must keep pushing the car, $N(\varphi) \geq 0$.

To compute N we measure φ from the bottom (the entrance) of the looping and use polar coordinates about its centre, with $\boldsymbol{e}_r$ pointing outwards. The normal

force points inwards, the radial component of gravity is $mg\cos\varphi$, and the radial acceleration is $-R\dot{\varphi}^2 = -v^2/R$. The radial momentum balance therefore reads

$$-N(\varphi) + mg\cos\varphi = -m\frac{v^2}{R} \qquad \Longrightarrow \qquad N(\varphi) = mg\cos\varphi + \frac{mv^2}{R}.$$

Inserting $v^2 = 2g(h_0 - h)$ with $h = R(1-\cos\varphi)$,

$$N(\varphi) = mg\left(3\cos\varphi - 2 + \frac{2h_0}{R}\right).$$

The normal force is smallest at the top, $\varphi = \pi$, where $N(\pi) = mg(2h_0/R - 5)$, and the condition $N(\pi) \geq 0$ gives

$$h_0 \geq \frac{5}{2}R.$$

The force is largest at the entrance $\varphi = 0$; for the minimal height $h_0 = 5R/2$ it is $N(0) = mg(3 - 2 + 5) = 6mg$. The passengers would feel six times their own weight – another reason why loopings are not built circular. ◀

Lecture 5
30.09.2026

The pendulum provides a good opportunity to compare the three balance laws on one and the same problem.

Example 1.17 *(Pendulum).* A mass m hangs on a massless string of length R from a fixed pivot B; φ denotes the angle from the downward vertical. We seek the angular acceleration $\ddot{\varphi}$. The forces are the string force $-F\boldsymbol{e}_r$ and the weight, which in the polar basis about B has the components $mg\cos\varphi\,\boldsymbol{e}_r - mg\sin\varphi\,\boldsymbol{e}_\varphi$.

Momentum balance. With $r = R$ the acceleration is $-R\dot{\varphi}^2\boldsymbol{e}_r + R\ddot{\varphi}\,\boldsymbol{e}_\varphi$, so

$$(-F + mg\cos\varphi)\,\boldsymbol{e}_r - mg\sin\varphi\,\boldsymbol{e}_\varphi = m(-R\dot{\varphi}^2\boldsymbol{e}_r + R\ddot{\varphi}\,\boldsymbol{e}_\varphi).$$

The $\boldsymbol{e}_\varphi$-component gives the equation of motion $\ddot{\varphi} = -(g/R)\sin\varphi$, and the $\boldsymbol{e}_r$-component the string force $F = mg\cos\varphi + mR\dot{\varphi}^2$; note the same structure as the normal force in the looping.

Angular momentum balance. About the fixed pivot the string force has no torque, and gravity has the lever arm $R\sin\varphi$. Thus $-mgR\sin\varphi = mR^2\ddot{\varphi}$, which is the same equation of motion, obtained without ever introducing the unknown string force.

Work–energy balance. The system is conservative (the string force is perpendicular to the motion). With the potential measured from the height of the pivot and the mass released from rest at height $-h_0$,

$$-mgh_0 = -mgR\cos\varphi + \frac{m}{2}R^2\dot{\varphi}^2.$$

The left-hand side is constant, so differentiating with respect to time gives $0 = mgR\sin\varphi\,\dot{\varphi} + mR^2\dot{\varphi}\ddot{\varphi}$, i.e. once more $\ddot{\varphi} = -(g/R)\sin\varphi$. For small angles, $\sin\varphi \approx \varphi$, this is the equation of a harmonic oscillator with angular frequency $\sqrt{g/R}$. ◀

1.2.4 Particle impact

We finally consider a particle hitting a rigid and smooth wall. "Rigid" means that the wall neither moves nor deforms; "smooth" means that it is frictionless and can only push along its normal $\boldsymbol{e}_n$. During the impact the contact force $\boldsymbol{F}(t)$ is very large but acts only for a very short time $[t_-, t_+]$, so that it is not the force itself that matters but its time integral.

Definition 1.15. The *impulse* of a force over the interval $[t_-, t_+]$ is $\hat{\boldsymbol{F}} = \int_{t_-}^{t_+} \boldsymbol{F}(t)\,\mathrm{d}t$ (unit N s).

Integrating the momentum balance over the impact gives $m\boldsymbol{v}(t_+) - m\boldsymbol{v}(t_-) = \hat{\boldsymbol{F}}$: the impulse is the total momentum transferred to the particle. It is helpful to split the impact into two phases at the instant t_0 of maximum compression. During the *compression* phase $[t_-, t_0]$ the particle is slowed down and deformed; during the *restitution* phase $[t_0, t_+]$ it (partially) regains its original shape and is pushed away. Denote the corresponding impulses by $\hat{F}_{\mathrm{comp}}\boldsymbol{e}_n$ and $\hat{F}_{\mathrm{rest}}\boldsymbol{e}_n$ – both are along $\boldsymbol{e}_n$, since the wall is smooth.

Decomposing the velocity into normal and tangential parts, $\boldsymbol{v} = v_n\boldsymbol{e}_n + v_t\boldsymbol{e}_t$, the momentum balance for the two phases reads

$$m\boldsymbol{v}(t_0) - m\boldsymbol{v}(t_-) = \hat{F}_{\mathrm{comp}}\boldsymbol{e}_n, \qquad m\boldsymbol{v}(t_+) - m\boldsymbol{v}(t_0) = \hat{F}_{\mathrm{rest}}\boldsymbol{e}_n.$$

The tangential components receive no impulse and are therefore unchanged,

$$v_t(t_-) = v_t(t_0) = v_t(t_+). \tag{1.22}$$

At the instant of maximum compression the normal velocity vanishes, $v_n(t_0) = 0$, so the normal components give $-mv_n(t_-) = \hat{F}_{\mathrm{comp}}$ and $mv_n(t_+) = \hat{F}_{\mathrm{rest}}$. These are two equations for three unknowns. The missing information is a property of the material: how much of the compression is recovered.

Definition 1.16. The *coefficient of restitution* is $e = \hat{F}_{\mathrm{rest}}/\hat{F}_{\mathrm{comp}} \in [0, 1]$. The impact is called *elastic* if $e = 1$ (full recovery), *plastic* if $e = 0$ (no recovery), and partially elastic otherwise.

Dividing the two normal equations by each other, $-v_n(t_+)/v_n(t_-) = e$, that is,

$$v_n(t_+) = -e\,v_n(t_-). \tag{1.23}$$

In an elastic impact the normal velocity is simply reversed; in a plastic impact the particle stays at the wall and slides along it with its unchanged tangential velocity; in between, the normal speed is reduced by the factor e. That e cannot exceed one follows from energy considerations: the kinetic energies before and after are $\frac{m}{2}(v_n^2 + v_t^2)$ and $\frac{m}{2}(e^2v_n^2 + v_t^2)$, and for $e > 1$ the passive wall would create energy out of nothing.

Example 1.26 *(Measuring the coefficient of restitution).* A particle is dropped from rest from height h_- onto the ground and rebounds to height h_+. The work–energy balance does not hold *across* the impact, where energy is lost, but it does hold during the free flights before and after. Hence $mgh_- = \frac{m}{2}v_n^2(t_-)$ and $mgh_+ = \frac{m}{2}v_n^2(t_+)$, and

$$\frac{h_+}{h_-} = \frac{v_n^2(t_+)}{v_n^2(t_-)} = e^2 \qquad \Longrightarrow \qquad e = \sqrt{h_+/h_-}.$$

Alternatively, one can measure the times between subsequent impacts. After a bounce with upward speed v_0, $x_2(t) = v_0t - \frac{1}{2}gt^2$, so the flight lasts $t^* = 2v_0/g$. The next flight starts with ev_0 and lasts $2ev_0/g$; hence $e = t_2^*/t_1^*$, the ratio of two consecutive flight times. ◀

Chapter 2

Dynamics of Systems of Particles

Lecture 6
05.10.2026

We now consider n particles with masses m_i and positions $\boldsymbol{r}_i$, $i = 1, \dots, n$, which may interact with each other. The goal of this chapter is to find balance laws for the system as a whole. The key observation will be that, thanks to Newton's third axiom, the internal forces cancel from the momentum and angular momentum balances, so that the motion of the system as a whole is governed by the external forces alone.

2.1 Kinematics

Definition 2.1. The *total mass* and the *centre of mass* of the system are

$$M = \sum_{i=1}^{n} m_i, \qquad \boldsymbol{r}_{\mathrm{cm}} = \frac{1}{M} \sum_{i=1}^{n} m_i \boldsymbol{r}_i. \tag{2.1}$$

The particles of a system are frequently connected to each other or to the surroundings. Such connections are described by constraints, and the most important ones can be written as equations between the positions.

Definition 2.2. A constraint is *holonomic* if it can be written as an equation between the positions (and time),

$$f(\boldsymbol{r}_1, \dots, \boldsymbol{r}_n, t) = 0.$$

A rigid link between particles i and j, which cannot change its length, is described by the holonomic constraint $|\boldsymbol{r}_i(t) - \boldsymbol{r}_j(t)| = \text{const}$, and a rigid system by the set of constraints $|\boldsymbol{r}_i - \boldsymbol{r}_j| = \text{const}$ for all pairs i, j. Each particle contributes d degrees of freedom, and each independent constraint removes one:

Proposition 2.3. *A system of n particles in d dimensions with r independent holonomic constraints has $nd - r$ degrees of freedom.*

Example 2.1 *(Counting degrees of freedom).* Two particles connected by a rigid link in three dimensions have $2 \cdot 3 - 1 = 5$ DOFs: three for the position of one of them and two angles for the direction of the link. Two masses suspended from ropes wound around a stepped pulley with radii r_0 and r_1 move by $x_0 = r_0\varphi$ and $x_1 = r_1\varphi$; the whole system has a single DOF, the pulley angle φ. As a last example take four particles in the plane forming a quadrilateral of four rigid links (a diagonal spring is *not* a constraint), one corner pinned and one on a roller. They have $4 \cdot 2 = 8$ coordinates; the pin removes two, the roller one, and the links four, leaving $8 - 7 = 1$ DOF. ◀

2.2 Kinetics

Each particle i experiences an external force $\boldsymbol{F}_{i,\mathrm{ext}}$ and internal forces $\boldsymbol{F}_{ij}$ exerted by the other particles j. By Newton's third axiom $\boldsymbol{F}_{ij} = -\boldsymbol{F}_{ji}$, and both act along the

line connecting the particles. The net force on particle i is $\boldsymbol{F}_i = \boldsymbol{F}_{i,\text{ext}} + \sum_{j\neq i} \boldsymbol{F}_{ij}$, and when we sum over all particles the internal forces cancel pairwise:

$$\sum_{i=1}^{n} \boldsymbol{F}_i = \sum_{i=1}^{n} \boldsymbol{F}_{i,\text{ext}}. \tag{2.2}$$

2.2.1 Balance of linear momentum

Summing the momentum balances $\boldsymbol{F}_i = \frac{\mathrm{d}}{\mathrm{d}t}(m_i \boldsymbol{v}_i)$ of the individual particles and using (2.2) and the definition of the centre of mass,

$$\sum_{i=1}^{n} \boldsymbol{F}_{i,\text{ext}} = \frac{\mathrm{d}}{\mathrm{d}t} \sum_{i=1}^{n} m_i \boldsymbol{v}_i = \frac{\mathrm{d}}{\mathrm{d}t}\Big(M \, \frac{1}{M} \sum_{i=1}^{n} m_i \dot{\boldsymbol{r}}_i \Big) = \frac{\mathrm{d}}{\mathrm{d}t}(M \boldsymbol{v}_{\text{cm}}).$$

Theorem 2.4. *The total momentum of a system of particles is $\boldsymbol{P} = M\boldsymbol{v}_{\text{cm}}$, and it obeys*

$$\sum_{i=1}^{n} \boldsymbol{F}_{i,\text{ext}} = \frac{\mathrm{d}}{\mathrm{d}t}(M\boldsymbol{v}_{\text{cm}}), \qquad \textit{so that} \quad \sum_{i=1}^{n} \boldsymbol{F}_{i,\text{ext}} = M\boldsymbol{a}_{\text{cm}} \quad \textit{if } M \textit{ is constant.} \tag{2.3}$$

In words: the centre of mass moves like a single particle carrying the total mass, acted upon by the total external force. No matter how complicated the internal interactions are, they cannot move the centre of mass.

Example 2.4 *(Falling dumbbell).* Two masses m_1 and m_2 connected by a rigid massless rod are thrown into the air. The only external force is gravity, so $(m_1 + m_2)\boldsymbol{a}_{\text{cm}} = (m_1 + m_2)\boldsymbol{g}$, and the centre of mass moves on the parabola $\boldsymbol{r}_{\text{cm}}(t) = \frac{1}{2}\boldsymbol{g}t^2 + \boldsymbol{v}_{\text{cm}}(0)t + \boldsymbol{r}_{\text{cm}}(0)$ – regardless of how the dumbbell tumbles. To learn about the internal force, the rod force, we have to apply the momentum balance to the individual particles: $m_1\boldsymbol{a}_1 = m_1\boldsymbol{g} + \boldsymbol{F}_{12}$ and $m_2\boldsymbol{a}_2 = m_2\boldsymbol{g} + \boldsymbol{F}_{21}$ with $\boldsymbol{F}_{21} = -\boldsymbol{F}_{12}$. ◀

2.2.2 Work–energy balance

Summing the work–energy balances $T_i(t_2) - T_i(t_1) = W_{i,12}$ of the individual particles gives the work–energy balance of the system,

$$T(t_2) - T(t_1) = W_{12}, \qquad T = \frac{1}{2}\sum_{i=1}^{n} m_i |\boldsymbol{v}_i|^2, \qquad W_{12} = \sum_{i=1}^{n} \int_{\boldsymbol{r}_i(t_1)}^{\boldsymbol{r}_i(t_2)} \boldsymbol{F}_i \cdot \mathrm{d}\boldsymbol{r}_i. \tag{2.4}$$

There is an important difference to the momentum balance: *internal forces may do work*. The two forces of a compressed spring between two particles are equal and opposite, but the particles move differently, and the work of the two forces does not cancel. Accordingly, $\boldsymbol{F}_i$ here contains the internal forces. As before, conservative forces can be accounted for by a potential, and for a conservative system $T + V = \text{const}$.

There is, however, an important class of internal forces that do no work.

Proposition 2.5. *A rigid link never performs work.*

Proof. Differentiating the constraint $|\boldsymbol{r}_i - \boldsymbol{r}_j|^2 = \text{const}$ gives $2(\boldsymbol{r}_i - \boldsymbol{r}_j)\cdot(\boldsymbol{v}_i - \boldsymbol{v}_j) = 0$. The link forces act along the link, $\boldsymbol{F}_{ij} \parallel \boldsymbol{r}_i - \boldsymbol{r}_j$, and $\boldsymbol{F}_{ji} = -\boldsymbol{F}_{ij}$, so their combined work is

$$\int \boldsymbol{F}_{ij} \cdot \mathrm{d}\boldsymbol{r}_i + \int \boldsymbol{F}_{ji} \cdot \mathrm{d}\boldsymbol{r}_j = \int_{t_1}^{t_2} \boldsymbol{F}_{ij} \cdot (\boldsymbol{v}_i - \boldsymbol{v}_j)\,\mathrm{d}t = 0. \qquad \square$$

2.2.3 Balance of angular momentum

The angular momentum of the system about a point B is the sum of the individual angular momenta, $\boldsymbol{H}_B = \sum_i \boldsymbol{r}_{BP_i} \times \boldsymbol{p}_i$. Repeating the derivation of (1.16),

$$\dot{\boldsymbol{H}}_B = \sum_i (\boldsymbol{v}_{P_i} - \boldsymbol{v}_B) \times \boldsymbol{p}_i + \sum_i \boldsymbol{r}_{BP_i} \times \boldsymbol{F}_i = -\boldsymbol{v}_B \times \boldsymbol{P} + \sum_i \boldsymbol{r}_{BP_i} \times \boldsymbol{F}_i,$$

since $\boldsymbol{v}_{P_i} \times \boldsymbol{p}_i = \boldsymbol{0}$ and $\sum_i \boldsymbol{p}_i = \boldsymbol{P}$. The internal forces come in pairs that are equal, opposite and collinear, so their torques cancel, and only the torque $\boldsymbol{M}_B$ of the external forces remains.

Theorem 2.6. *For a system of particles, $\boldsymbol{M}_B = \dot{\boldsymbol{H}}_B + \boldsymbol{v}_B \times \boldsymbol{P}$ with $\boldsymbol{P} = M\boldsymbol{v}_{\mathrm{cm}}$, where $\boldsymbol{M}_B$ is the net external torque about B.*

Example 2.5 *(Particles on a disk).* Particles m_i are attached at radii r_i to a massless disk which rotates by the angle $\varphi(t)$ about its fixed centre B. Each particle moves on a circle, $\boldsymbol{v}_i = r_i\dot{\varphi}\,\boldsymbol{e}_{\varphi,i}$, so

$$\boldsymbol{H}_B = \sum_{i=1}^{n} r_i \boldsymbol{e}_{r,i} \times m_i r_i \dot{\varphi}\,\boldsymbol{e}_{\varphi,i} = \Big(\sum_{i=1}^{n} m_i r_i^2\Big)\dot{\varphi}\,\boldsymbol{e}_3,$$

and the angular momentum balance about the fixed point B gives $M_B^{\mathrm{ext}} = I_B\ddot{\varphi}$ with $I_B = \sum_i m_i r_i^2$. If the external torque is due to gravity, it can be simplified: $\boldsymbol{M}_B^{\mathrm{ext}} = \sum_i \boldsymbol{r}_{BP_i} \times m_i\boldsymbol{g} = \big(\sum_i m_i \boldsymbol{r}_{BP_i}\big) \times \boldsymbol{g} = M\boldsymbol{r}_{B,\mathrm{cm}} \times \boldsymbol{g}$. In a homogeneous gravitational field, the weight acts as if the total mass were concentrated at the centre of mass. ◀

Example 2.6 *(A ring of particles).* For n particles of mass M/n evenly distributed on a circle of radius R, $I_B = \sum_i (M/n)R^2 = MR^2$. In the limit $n \to \infty$ this is the moment of inertia of a thin ring about its centre. ◀

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Example 2.8 *(Atwood machine).* Two masses m_1 (left) and m_2 (right) hang from a rope passing over a massless wheel of radius R, whose bearing B is fixed; the wheel is driven by a torque M_T. We seek the acceleration and the bearing force.

If x_1 is the downward displacement of m_1 and x_2 the upward displacement of m_2, then $x_1 = R\varphi = x_2 =: x$; the system has one DOF. We cut the rope on both sides of the wheel and call the rope forces S_1 and S_2. The momentum balances of the two masses are

$$m_1\ddot{x} = m_1 g - S_1, \qquad m_2\ddot{x} = -m_2 g + S_2.$$

The wheel is massless, so the forces and torques on it balance: the bearing force is $B_2 = S_1 + S_2$ (vertically) and $B_1 = 0$, and the angular momentum balance about B reads $0 = I_B\ddot{\varphi} = S_1 R - S_2 R + M_T$. Adding the two momentum balances and using $S_2 - S_1 = M_T/R$,

$$(m_1 + m_2)\ddot{x} = (m_1 - m_2)g + \frac{M_T}{R} \qquad \Longrightarrow \qquad \ddot{x} = \frac{m_1 - m_2}{m_1 + m_2}\, g + \frac{M_T}{(m_1 + m_2)R}. \quad (*)$$

The bearing force is

$$B_2 = S_1 + S_2 = (m_1 + m_2)g + (m_2 - m_1)\ddot{x}.$$

The first term is the static value; the second appears only because the system moves. In particular, the static result $B_2 = (m_1 + m_2)g$ is wrong in dynamics.

The same result follows from the work–energy balance, once we know how much work a torque does. A torque M_T is equivalent to a couple of forces F applied at distance R on either side of the axis, $M_T = 2FR$. When the wheel turns by $\mathrm{d}\varphi$ each force moves by $R\,\mathrm{d}\varphi$, so the work is $2FR\,\mathrm{d}\varphi = M_T\,\mathrm{d}\varphi$. Hence

$$\frac{m_1 + m_2}{2}\dot{x}^2 + m_2 g x - m_1 g x - [\text{initial value}] = M_T(\varphi - \varphi_0),$$

and differentiating with respect to time, with $\dot{\varphi} = \dot{x}/R$, gives $(m_1 + m_2)\dot{x}\ddot{x} + (m_2 - m_1)g\dot{x} = M_T\dot{x}/R$, i.e. again $(*)$.

Finally, we may also treat the two masses together with the wheel as one system. Placing the origin at the axle, the masses hang at horizontal positions $\mp R$ and their heights change by $\mp x$, so $\boldsymbol{a}_{\mathrm{cm}} = \frac{m_2 - m_1}{m_1 + m_2}\ddot{x}\,\boldsymbol{e}_2$. The momentum balance of the whole system, $(m_1 + m_2)\boldsymbol{a}_{\mathrm{cm}} = B_1\boldsymbol{e}_1 + (B_2 - m_1 g - m_2 g)\boldsymbol{e}_2$, reproduces $B_1 = 0$ and the expression for B_2. ◀

2.3 Collisions and variable mass

2.3.1 Particle collisions

Two smooth spheres collide. The plane tangent to both at the point of contact is the *collision plane*; let $\boldsymbol{e}_n$ be its normal and $\boldsymbol{e}_t$ a direction in it. Since the spheres are smooth, the contact impulses are normal: particle 2 receives $\hat{F}\boldsymbol{e}_n$ and particle 1, by Newton's third axiom, $-\hat{F}\boldsymbol{e}_n$:

$$m_1\boldsymbol{v}_1(t_+) - m_1\boldsymbol{v}_1(t_-) = -\hat{F}\boldsymbol{e}_n, \qquad m_2\boldsymbol{v}_2(t_+) - m_2\boldsymbol{v}_2(t_-) = \hat{F}\boldsymbol{e}_n.$$

As for the wall, the tangential velocity components are unchanged, $v_i^t(t_+) = v_i^t(t_-)$. For the normal components we again split the impact into compression and restitution:

$$\begin{aligned} m_1 v_1^n(t_0) - m_1 v_1^n(t_-) &= -\hat{F}_{\mathrm{comp}}, & m_2 v_2^n(t_0) - m_2 v_2^n(t_-) &= \hat{F}_{\mathrm{comp}}, \\ m_1 v_1^n(t_+) - m_1 v_1^n(t_0) &= -\hat{F}_{\mathrm{rest}}, & m_2 v_2^n(t_+) - m_2 v_2^n(t_0) &= \hat{F}_{\mathrm{rest}}. \end{aligned}$$

These are four equations for six unknowns (everything except the initial velocities). The two missing equations are the definition of the coefficient of restitution, $e = \hat{F}_{\mathrm{rest}}/\hat{F}_{\mathrm{comp}}$, and the condition that at the instant of maximum compression both bodies have the same normal velocity, $v_1^n(t_0) = v_2^n(t_0) =: v_0$.

To solve, we add the two compression equations; $\hat{F}_{\mathrm{comp}}$ drops out, and $v_0 = \big(m_1 v_1^n(t_-) + m_2 v_2^n(t_-)\big)/(m_1 + m_2)$ turns out to be the normal velocity of the centre of mass. Then $\hat{F}_{\mathrm{comp}} = m_2\big(v_0 - v_2^n(t_-)\big) = \frac{m_1 m_2}{m_1 + m_2}\big(v_1^n(t_-) - v_2^n(t_-)\big)$, the total normal impulse is $(1 + e)\hat{F}_{\mathrm{comp}}$, and we obtain

$$v_1^n(t_+) = \frac{m_1 - e\,m_2}{m_1 + m_2}\,v_1^n(t_-) + \frac{(1 + e)\,m_2}{m_1 + m_2}\,v_2^n(t_-), \tag{2.5}$$

$$v_2^n(t_+) = \frac{m_2 - e\,m_1}{m_1 + m_2}\,v_2^n(t_-) + \frac{(1 + e)\,m_1}{m_1 + m_2}\,v_1^n(t_-). \tag{2.6}$$

Subtracting these two expressions shows that the coefficient of restitution is the ratio of the relative normal velocities after and before the collision,

$$e = -\frac{v_2^n(t_+) - v_1^n(t_+)}{v_2^n(t_-) - v_1^n(t_-)}. \tag{2.7}$$

Moreover, since there are no external forces in the normal direction, the momentum is conserved, $m_1 v_1^n(t_+) + m_2 v_2^n(t_+) = m_1 v_1^n(t_-) + m_2 v_2^n(t_-)$.

Three special cases are worth remembering (Examples 2.9 and 2.10). Letting $m_2 \to \infty$ with $v_2^n = 0$ we recover the rigid wall, $v_1^n(t_+) = -e v_1^n(t_-)$. In a plastic collision, $e = 0$, the two bodies move on together with the velocity of the centre of mass. And in an elastic collision of equal masses, $e = 1$ and $m_1 = m_2$, the two particles simply exchange their normal velocities – the principle of Newton's cradle.

2.3.2 Particles with variable mass

A body that gains or loses mass – a rocket burning fuel, a raindrop collecting moisture – requires care, because the momentum balance applies to a *fixed* collection of matter. Consider a body of mass $m(t)$ and velocity $\boldsymbol{v}(t)$ which, during the time $\mathrm{d}t$, absorbs a mass $\mathrm{d}m$ arriving with the velocity $\boldsymbol{v}_m$. We apply the momentum balance to the closed system consisting of the body and the mass element. At time t its momentum is $\boldsymbol{p}(t) = m\boldsymbol{v} + \mathrm{d}m\,\boldsymbol{v}_m$, and at time $t + \mathrm{d}t$, using Taylor expansions,

$$\boldsymbol{p}(t+\mathrm{d}t) = [m + \dot{m}\,\mathrm{d}t + O(\mathrm{d}t^2)]\,[\boldsymbol{v} + \dot{\boldsymbol{v}}\,\mathrm{d}t + O(\mathrm{d}t^2)] = m\boldsymbol{v} + \dot{m}\boldsymbol{v}\,\mathrm{d}t + m\dot{\boldsymbol{v}}\,\mathrm{d}t + O(\mathrm{d}t^2).$$

With $\mathrm{d}m = \dot{m}\,\mathrm{d}t$ the rate of change of momentum is

$$\dot{\boldsymbol{p}} = \lim_{\mathrm{d}t\to 0} \frac{\boldsymbol{p}(t+\mathrm{d}t) - \boldsymbol{p}(t)}{\mathrm{d}t} = \dot{m}\boldsymbol{v} + m\dot{\boldsymbol{v}} - \dot{m}\boldsymbol{v}_m.$$

Theorem 2.7 (Rocket equation)**.** *For a body of variable mass,*

$$\sum_i \boldsymbol{F}_i = m\boldsymbol{a} + \dot{m}(\boldsymbol{v} - \boldsymbol{v}_m) = m\boldsymbol{a} - \dot{m}\,\boldsymbol{v}_{rel}, \qquad \textit{equivalently} \qquad m\boldsymbol{a} = \sum_i \boldsymbol{F}_i + \dot{m}\,\boldsymbol{v}_{rel}, \tag{2.8}$$

where $\boldsymbol{v}_{rel} = \boldsymbol{v}_m - \boldsymbol{v}$ is the velocity of the gained or ejected mass relative to the body, and $\dot{m}\,\boldsymbol{v}_{rel}$ is the thrust*.*

Note that the naive formula $\sum \boldsymbol{F} = \frac{\mathrm{d}}{\mathrm{d}t}(m\boldsymbol{v})$ is *not* correct for a body of variable mass: it would give different results in different inertial frames. Only the relative velocity of the mass flow matters.

Example 2.11 *(Rocket launch).* A rocket moves vertically upwards, $\boldsymbol{v} = \dot{x}\boldsymbol{e}_2$, and ejects its exhaust gas downwards with the velocity $\boldsymbol{v}_{\mathrm{rel}} = -c\boldsymbol{e}_2$ $(c > 0)$ relative to the rocket. It is subject to gravity and to a drag F_{drag}. The rocket equation gives

$$(-F_{\mathrm{drag}} - mg)\,\boldsymbol{e}_2 = m\ddot{x}\,\boldsymbol{e}_2 + \dot{m}c\,\boldsymbol{e}_2 \qquad \Longrightarrow \qquad m\ddot{x} = -F_{\mathrm{drag}} - mg - \dot{m}c.$$

Since the rocket loses mass, $\dot{m} < 0$, the last term $-\dot{m}c$ is positive; it is the thrust, and the rocket accelerates upwards only if it exceeds $mg + F_{\mathrm{drag}}$. ◀

Chapter 3

Dynamics of Rigid Bodies

Lecture 8
12.10.2026

A rigid body can be thought of as the limit of a system of particles, $n \to \infty$ with $\Delta m_i \to 0$, in which all mutual distances are fixed: $|\boldsymbol{r}_{AB}| = \text{const}$ for any two points A, B of the body $\mathcal{B}$. Everything we derived for systems of particles therefore carries over, with sums replaced by integrals. What is new is that a rigid body has an orientation, and its rotation must be described and accounted for. In the kinematics this leads to the angular velocity vector and the transfer formulas; in the kinetics it leads to the moment of inertia tensor.

3.1 Kinematics

With the density $\rho = \mathrm{d}m/\mathrm{d}V$, the total mass and the centre of mass of a body are

$$M = \lim \sum_{i=1}^{n} \Delta m_i = \int_{\mathcal{B}} \mathrm{d}m = \int_{\mathcal{B}} \rho \, \mathrm{d}V, \qquad \boldsymbol{r}_{\mathrm{cm}} = \frac{1}{M} \int_{\mathcal{B}} \boldsymbol{r} \, \mathrm{d}m. \tag{3.1}$$

How many degrees of freedom does a rigid body have? In the plane its position is fixed by the two coordinates of one of its points and one angle; in space by three coordinates and three angles.

Proposition 3.1. *An unconstrained rigid body in d dimensions has $d(d+1)/2$ degrees of freedom: d translations and $d(d-1)/2$ rotations (1 DOF in 1D, 3 in 2D, 6 in 3D). Consequently n rigid bodies with k independent holonomic constraints have $n\,d(d+1)/2 - k$ DOFs.*

3.1.1 Velocity and acceleration of points of a rigid body

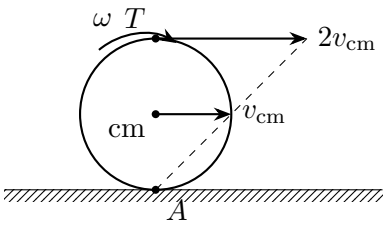
For two points A and B of the body we have $\boldsymbol{r}_B = \boldsymbol{r}_A + \boldsymbol{r}_{AB}$, hence $\boldsymbol{v}_B = \boldsymbol{v}_A + \dot{\boldsymbol{r}}_{AB}$ and $\boldsymbol{a}_B = \boldsymbol{a}_A + \ddot{\boldsymbol{r}}_{AB}$. Since the length of $\boldsymbol{r}_{AB}$ is constant, the vector can only change its direction; its derivatives must therefore be expressible through the rotation of the body (and vanish if the body does not rotate).

To make this precise, recall from Mechanics I the angular velocity vector $\boldsymbol{\omega}(t) = \omega(t)\,\boldsymbol{e}_\omega(t)$, whose direction $\boldsymbol{e}_\omega$ is the axis of rotation; in plane motion $\boldsymbol{e}_\omega = \boldsymbol{e}_3$ is constant and points out of the plane. If the body rotates about a fixed point Π, every point moves according to $\boldsymbol{v} = \boldsymbol{\omega} \times \boldsymbol{r}_{\Pi P}$. Subtracting this relation for two points A and B, $\boldsymbol{v}_B - \boldsymbol{v}_A = \boldsymbol{\omega} \times (\boldsymbol{r}_{\Pi B} - \boldsymbol{r}_{\Pi A}) = \boldsymbol{\omega} \times \boldsymbol{r}_{AB}$. The result does not refer to Π any more and holds for general rigid motion.

Theorem 3.2 (Velocity transfer formula)**.** *For any two points A and B of a rigid body,*

$$\boldsymbol{v}_B = \boldsymbol{v}_A + \boldsymbol{\omega} \times \boldsymbol{r}_{AB}. \tag{3.2}$$

Example 3.1 *(Rolling disk).*



A disk of radius R rolls without slipping. Rolling means that the distance travelled by the centre equals the arc length unrolled, $x_{\mathrm{cm}} = R\varphi$, so $v_{\mathrm{cm}} = R\omega$. For rolling to the right the disk turns clockwise, $\boldsymbol{\omega} = -\omega \boldsymbol{e}_3$, and the transfer formula gives for the top point T and the contact point A

$$\begin{aligned} \boldsymbol{v}_T &= R\omega \boldsymbol{e}_1 + (-\omega \boldsymbol{e}_3) \times R\boldsymbol{e}_2 = 2R\omega \boldsymbol{e}_1, \\ \boldsymbol{v}_A &= R\omega \boldsymbol{e}_1 + (-\omega \boldsymbol{e}_3) \times (-R\boldsymbol{e}_2) = \boldsymbol{0}. \end{aligned}$$

The contact point is momentarily at rest, and the velocities along the vertical diameter grow linearly from 0 to $2v_{\mathrm{cm}}$, exactly as if the disk were rotating about A. ◀

The last observation is general. In plane motion with $\omega \neq 0$ there is always a point Π of the body (or of its rigid extension) with $\boldsymbol{v}_\Pi = \boldsymbol{0}$, the *instantaneous centre of rotation* (ICR), and the body momentarily rotates about it: $\boldsymbol{v}_P = \boldsymbol{\omega} \times \boldsymbol{r}_{\Pi P}$ for every point P. If the velocity $\boldsymbol{v}_A$ of one point A is known, the ICR is found at

$$\boldsymbol{r}_{A\Pi} = \frac{1}{\omega}\, \boldsymbol{e}_\omega \times \boldsymbol{v}_A, \tag{3.3}$$

on the line through A perpendicular to $\boldsymbol{v}_A$, at the distance $|\boldsymbol{v}_A|/\omega$. Indeed, by (3.2), $\boldsymbol{v}_\Pi = \boldsymbol{v}_A + \boldsymbol{e}_\omega \times (\boldsymbol{e}_\omega \times \boldsymbol{v}_A) = \boldsymbol{v}_A - \boldsymbol{v}_A = \boldsymbol{0}$, since $\boldsymbol{e}_\omega \times (\boldsymbol{e}_\omega \times \boldsymbol{v}) = -\boldsymbol{v}$ for any $\boldsymbol{v}$ in the plane.

A second useful consequence of rigidity is obtained by differentiating $\boldsymbol{r}_{AB} \cdot \boldsymbol{r}_{AB} = \text{const}$:

$$2\, \dot{\boldsymbol{r}}_{AB} \cdot \boldsymbol{r}_{AB} = 0 \quad \Longleftrightarrow \quad (\boldsymbol{v}_B - \boldsymbol{v}_A) \cdot \boldsymbol{r}_{AB} = 0.$$

Proposition 3.3 (Theorem of projected velocities)**.** *For any two points of a rigid body, $\boldsymbol{v}_A \cdot \boldsymbol{r}_{AB} = \boldsymbol{v}_B \cdot \boldsymbol{r}_{AB}$: their velocity components along the connecting line are equal.*

Example 3.2 *(Slider–crank mechanism).* A crank AB of length L turns about the fixed pivot A with angular velocity $\boldsymbol{\omega}_2$ and makes the angle α with the ground. The connecting rod BC of length $2L$, inclined at the angle β, drives a slider C along the ground, $\boldsymbol{v}_C = -v_C \boldsymbol{e}_1$. Given ω_2, we want v_C.

With $\boldsymbol{e}_1$ to the right and $\boldsymbol{e}_2$ upwards the geometry is $\boldsymbol{r}_{AB} = -L\cos\alpha\, \boldsymbol{e}_1 + L\sin\alpha\, \boldsymbol{e}_2$ and $\boldsymbol{r}_{BC} = -2L\cos\beta\, \boldsymbol{e}_1 - 2L\sin\beta\, \boldsymbol{e}_2$, where $L\sin\alpha = 2L\sin\beta$ because C and A are at the same height. Since A is fixed and $\boldsymbol{\omega}_2 = \omega_2 \boldsymbol{e}_3$, the velocity of B is

$$\boldsymbol{v}_B = \boldsymbol{\omega}_2 \times \boldsymbol{r}_{AB} = -L\omega_2 \big(\sin\alpha\, \boldsymbol{e}_1 + \cos\alpha\, \boldsymbol{e}_2 \big).$$

The quickest route to v_C is the theorem of projected velocities applied to the rod, $\boldsymbol{v}_B \cdot \boldsymbol{r}_{BC} = \boldsymbol{v}_C \cdot \boldsymbol{r}_{BC}$:

$$2L^2\omega_2 \big(\sin\alpha\cos\beta + \cos\alpha\sin\beta \big) = 2L v_C \cos\beta$$

and therefore

$$v_C = \omega_2 L\big(\sin\alpha + \cos\alpha\tan\beta\big) = \omega_2 L\,\frac{\sin(\alpha+\beta)}{\cos\beta}.$$

Alternatively one can use the transfer formula twice, $\boldsymbol{v}_C = \boldsymbol{v}_A + \boldsymbol{\omega}_2\times\boldsymbol{r}_{AB} + \boldsymbol{\omega}_1\times\boldsymbol{r}_{BC} = -v_C\boldsymbol{e}_1$, which gives two scalar equations for v_C and the angular velocity $\boldsymbol{\omega}_1$ of the rod. In terms of the angles, $\boldsymbol{\omega}_1 = \dot\beta\,\boldsymbol{e}_3$ and $\boldsymbol{\omega}_2 = -\dot\alpha\,\boldsymbol{e}_3$. ◀

Differentiating the velocity transfer formula with respect to time, and using $\dot{\boldsymbol{r}}_{AB} = \boldsymbol{\omega}\times\boldsymbol{r}_{AB}$, gives the corresponding relation for accelerations.

Theorem 3.4 (Acceleration transfer formula)**.**

$$\boldsymbol{a}_B = \boldsymbol{a}_A + \dot{\boldsymbol{\omega}}\times\boldsymbol{r}_{AB} + \boldsymbol{\omega}\times(\boldsymbol{\omega}\times\boldsymbol{r}_{AB}). \tag{3.4}$$

In plane motion the last term simplifies to the centripetal term $-\omega^2\boldsymbol{r}_{AB}$.

Example 3.3 *(Rolling disk, accelerations).* With $\boldsymbol{a}_{\mathrm{cm}} = R\dot\omega\boldsymbol{e}_1$ and $\dot{\boldsymbol{\omega}} = -\dot\omega\boldsymbol{e}_3$,

$$\boldsymbol{a}_T = R\dot\omega\boldsymbol{e}_1 + (-\dot\omega\boldsymbol{e}_3)\times R\boldsymbol{e}_2 - \omega^2 R\boldsymbol{e}_2 = 2R\dot\omega\boldsymbol{e}_1 - R\omega^2\boldsymbol{e}_2,$$
$$\boldsymbol{a}_A = R\dot\omega\boldsymbol{e}_1 + (-\dot\omega\boldsymbol{e}_3)\times(-R\boldsymbol{e}_2) - \omega^2(-R\boldsymbol{e}_2) = R\omega^2\boldsymbol{e}_2.$$

The contact point, although momentarily at rest, is accelerated upwards towards the centre. The ICR has zero velocity, but in general *not* zero acceleration – a fact that has to be kept in mind whenever the ICR is used in kinetics. ◀

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14.10.2026

Example 3.5 *(Ladder sliding down a wall).* Two blocks of mass m are connected by a massless rigid rod of length L; block A slides on a frictionless vertical wall and block B on the frictionless floor. Let φ be the angle of the rod with the vertical. The system is released from rest from (almost) vertical. We want the velocities, accelerations and forces.

Two particles in the plane have four coordinates; the wall, the floor and the rod impose three constraints, so there is a single DOF, $\varphi(t)$. With $\boldsymbol{r}_A = L\cos\varphi\,\boldsymbol{e}_2$ and $\boldsymbol{r}_B = L\sin\varphi\,\boldsymbol{e}_1$,

$$\boldsymbol{v}_A = -L\dot\varphi\sin\varphi\,\boldsymbol{e}_2, \qquad \boldsymbol{v}_B = L\dot\varphi\cos\varphi\,\boldsymbol{e}_1.$$

The system is conservative, so energy conservation (potential zero at floor level; only A changes height) gives

$$mgL = mgL\cos\varphi + \frac{m}{2}(v_A^2 + v_B^2) = mgL\cos\varphi + \frac{m}{2}L^2\dot\varphi^2,$$

hence

$$\dot\varphi^2 = \frac{2g}{L}\,(1-\cos\varphi),$$

and differentiating in time, $\ddot\varphi = (g/L)\sin\varphi$.

Differentiating the velocities (or, equivalently, using the acceleration transfer formula with $\boldsymbol{r}_{AB} = L\sin\varphi\,\boldsymbol{e}_1 - L\cos\varphi\,\boldsymbol{e}_2$) gives the downward acceleration of A and the horizontal acceleration of B:

$$\begin{aligned} a_A &= L\ddot\varphi\sin\varphi + L\dot\varphi^2\cos\varphi = g\sin^2\varphi + 2g(1-\cos\varphi)\cos\varphi \\ &= g(1 + 2\cos\varphi - 3\cos^2\varphi), \\ a_B &= L\ddot\varphi\cos\varphi - L\dot\varphi^2\sin\varphi = g\sin\varphi\cos\varphi - 2g(1-\cos\varphi)\sin\varphi \\ &= g\sin\varphi\,(3\cos\varphi - 2). \end{aligned}$$

To find the forces, let S be the rod force (positive in tension), H the force of the wall on A and V the force of the floor on B. The horizontal momentum balance of B, in which V does not appear, and the vertical one of A read

$$ma_B = -S\sin\varphi, \qquad ma_A = mg + S\cos\varphi \qquad \Longrightarrow \qquad S = mg\,(2 - 3\cos\varphi),$$

and the remaining balances give $V = mg - S\cos\varphi$ and $H = -S\sin\varphi$.

The result tells an interesting story. As long as $\cos\varphi > 2/3$, $S < 0$: the rod is in compression and pushes A against the wall, $H > 0$. At $\cos\varphi = 2/3$, i.e. $\varphi \approx 48.2°$, the wall force vanishes; beyond that point the wall would have to pull on the block, which it cannot do. Block A separates from the wall and the above solution ceases to be valid. ◀

3.1.2 Rigid body kinematics in three dimensions

The velocity transfer formula holds unchanged in three dimensions, but now the axis of rotation $\boldsymbol{e}_\omega(t)$ may change its direction; at each instant it defines the *instantaneous axis of rotation* (IAR). In contrast to the ICR in the plane, the points on the IAR need not be at rest: they may move along the axis, $\boldsymbol{v}_{\mathrm{IAR}} = v_{\mathrm{IAR}}\,\boldsymbol{e}_\omega$, so that the general instantaneous motion is a screw. If, however, two points of a body are at rest, the IAR passes through both of them.

Example 3.6 *(Rolling cone).* A cone has its apex O fixed on a vertical axis at height R; its axis $\boldsymbol{e}_r$ is horizontal with length L, and its base circle of radius R rolls on the ground, touching it at A. The cone precesses about the vertical with the angle θ and spins about its own axis with the angle φ. Given $\dot\theta$, we seek $\boldsymbol{\omega}$.

For any point C of the cone, $\boldsymbol{v}_C = \boldsymbol{v}_O + \boldsymbol{\omega}\times\boldsymbol{r}_{OC} = \boldsymbol{\omega}\times\boldsymbol{r}_{OC}$, since O is fixed. Rolling means that the contact point is at rest, so $\boldsymbol{\omega}\times\boldsymbol{r}_{OA} = \boldsymbol{0}$, and as both vectors are non-zero, $\boldsymbol{\omega} \parallel \boldsymbol{r}_{OA}$: with two points at rest, the line OA *is* the IAR.

The angular velocity is the sum of the precession about the vertical and the spin about the cone axis, $\boldsymbol{\omega} = \dot\theta\,\boldsymbol{e}_3 - \dot\varphi\,\boldsymbol{e}_r$. Since $\boldsymbol{r}_{OA} = L\boldsymbol{e}_r - R\boldsymbol{e}_3$, parallelism requires $\dot\theta/\dot\varphi = R/L = \tan\alpha$, where α is the half-angle of the cone; this is the rolling condition $R\dot\varphi = L\dot\theta$. Therefore

$$\boldsymbol{\omega} = \dot\theta\,\boldsymbol{e}_3 - \frac{L}{R}\,\dot\theta\,\boldsymbol{e}_r.$$

The example illustrates that angular velocities are additive: the total rotation is the vector sum of the individual rotations. ◀

3.2 Kinetics

3.2.1 Balance of linear momentum

Passing to the limit in the momentum of a system of particles, the momentum of a rigid body is

$$\boldsymbol{P} = \lim\sum_{i=1}^{n}\boldsymbol{v}_i\Delta m_i = \int_{\mathcal{B}}\boldsymbol{v}\,\mathrm{d}m = M\boldsymbol{v}_{\mathrm{cm}},$$

where the last equality follows by differentiating $\boldsymbol{r}_{\mathrm{cm}} = \frac{1}{M}\int_{\mathcal{B}}\boldsymbol{r}\,\mathrm{d}m$. The momentum balance of the system carries over without change.

Theorem 3.5. $\dfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{P} = \sum_i \boldsymbol{F}_i^{\mathrm{ext}}$, *and for constant mass* $M\boldsymbol{a}_{\mathrm{cm}} = \sum_i \boldsymbol{F}_i^{\mathrm{ext}}$.

As far as its centre of mass is concerned, a rigid body is a "super-particle"; under gravity alone, for instance, $\boldsymbol{a}_{\mathrm{cm}} = \boldsymbol{g}$. The momentum balance, however, says nothing about the rotation. For that we need the angular momentum balance, which for a rigid body is a genuinely independent law.

3.2.2 Balance of angular momentum

The angular momentum balance for systems, $\boldsymbol{M}_B = \dot{\boldsymbol{H}}_B + \boldsymbol{v}_B \times \boldsymbol{P}$ with $\boldsymbol{P} = M\boldsymbol{v}_{\mathrm{cm}}$, also carries over. The work lies in computing the angular momentum of a rigid body about a point B of the body,

$$\boldsymbol{H}_B = \lim \sum_{i=1}^{n} \boldsymbol{r}_{BP_i} \times \Delta m_i \boldsymbol{v}_i = \int_{\mathcal{B}} (\boldsymbol{r} - \boldsymbol{r}_B) \times \boldsymbol{v}\, \mathrm{d}m.$$

The velocity of each point is given by the transfer formula, $\boldsymbol{v} = \boldsymbol{v}_B + \boldsymbol{\omega} \times (\boldsymbol{r} - \boldsymbol{r}_B)$. Inserting it, the integral splits into two parts. The first is

$$\int_{\mathcal{B}} (\boldsymbol{r} - \boldsymbol{r}_B)\, \mathrm{d}m \times \boldsymbol{v}_B = \Big[\int_{\mathcal{B}} \boldsymbol{r}\, \mathrm{d}m - \boldsymbol{r}_B \int_{\mathcal{B}} \mathrm{d}m \Big] \times \boldsymbol{v}_B = M(\boldsymbol{r}_{\mathrm{cm}} - \boldsymbol{r}_B) \times \boldsymbol{v}_B.$$

For the second, write $\bar{\boldsymbol{r}} = \boldsymbol{r} - \boldsymbol{r}_B$ and use $\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) = \boldsymbol{b}(\boldsymbol{a} \cdot \boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a} \cdot \boldsymbol{b})$:

$$\bar{\boldsymbol{r}} \times (\boldsymbol{\omega} \times \bar{\boldsymbol{r}}) = (\bar{\boldsymbol{r}} \cdot \bar{\boldsymbol{r}})\, \boldsymbol{\omega} - \bar{\boldsymbol{r}}\, (\bar{\boldsymbol{r}} \cdot \boldsymbol{\omega}) = \big[(\bar{\boldsymbol{r}} \cdot \bar{\boldsymbol{r}}) \underline{\underline{\mathbb{1}}} - \bar{\boldsymbol{r}} \bar{\boldsymbol{r}}^{\mathsf{T}} \big] \boldsymbol{\omega},$$

where $\underline{\underline{\mathbb{1}}}$ is the unit tensor and $\bar{\boldsymbol{r}}\bar{\boldsymbol{r}}^{\mathsf{T}}$ the dyadic product. The angular velocity is the same for all points of the body and can be taken out of the integral. What remains depends only on the distribution of mass in the body, and deserves a name.

Definition 3.6. The *(mass) moment of inertia tensor* of the body $\mathcal{B}$ with respect to the point B is

$$\underline{\underline{I}}_B = \int_{\mathcal{B}} \big[(\bar{\boldsymbol{r}} \cdot \bar{\boldsymbol{r}}) \underline{\underline{\mathbb{1}}} - \bar{\boldsymbol{r}} \bar{\boldsymbol{r}}^{\mathsf{T}} \big]\, \mathrm{d}m, \qquad \bar{\boldsymbol{r}} = \boldsymbol{r} - \boldsymbol{r}_B. \tag{3.5}$$

Theorem 3.7 (Angular momentum balance for a rigid body)**.** *The angular momentum of a rigid body about a point B of the body is* $\boldsymbol{H}_B = M(\boldsymbol{r}_{\mathrm{cm}} - \boldsymbol{r}_B) \times \boldsymbol{v}_B + \underline{\underline{I}}_B \boldsymbol{\omega}$, *and*

$$\boldsymbol{M}_B = \frac{\mathrm{d}}{\mathrm{d}t} \Big[M(\boldsymbol{r}_{\mathrm{cm}} - \boldsymbol{r}_B) \times \boldsymbol{v}_B + \underline{\underline{I}}_B \boldsymbol{\omega} \Big] + \boldsymbol{v}_B \times \boldsymbol{P}. \tag{3.6}$$

In components, with $\bar{\boldsymbol{r}} = (\bar{x}_1, \bar{x}_2, \bar{x}_3)$, we have $(\bar{\boldsymbol{r}} \cdot \bar{\boldsymbol{r}}) = \bar{x}_1^2 + \bar{x}_2^2 + \bar{x}_3^2$ and $(\bar{\boldsymbol{r}}\bar{\boldsymbol{r}}^{\mathsf{T}})_{ij} = \bar{x}_i \bar{x}_j$, so that

$$[\underline{\underline{I}}_B] = \int_{\mathcal{B}} \begin{bmatrix} \bar{x}_2^2 + \bar{x}_3^2 & -\bar{x}_1\bar{x}_2 & -\bar{x}_1\bar{x}_3 \\ -\bar{x}_1\bar{x}_2 & \bar{x}_1^2 + \bar{x}_3^2 & -\bar{x}_2\bar{x}_3 \\ -\bar{x}_1\bar{x}_3 & -\bar{x}_2\bar{x}_3 & \bar{x}_1^2 + \bar{x}_2^2 \end{bmatrix} \rho\, \mathrm{d}V. \tag{3.7}$$

The diagonal entries are the moments of inertia about the coordinate axes – integrals of the squared distance from the respective axis – and the off-diagonal entries are the products of inertia.

The general form of the balance is rarely needed, because a suitable choice of B makes it much simpler. If B is the centre of mass, the first term of $\boldsymbol{H}_B$ vanishes, and

so does $\boldsymbol{v}_B \times \boldsymbol{P} = \boldsymbol{v}_{\mathrm{cm}} \times M\boldsymbol{v}_{\mathrm{cm}}$; if B is a fixed point, $\boldsymbol{v}_B = \boldsymbol{0}$ and the same terms vanish. In both cases

$$\boldsymbol{M}_B = \frac{\mathrm{d}}{\mathrm{d}t}(\underline{\underline{I}}_B \boldsymbol{\omega}), \tag{3.8}$$

and one should always choose B in one of these two ways. If, in addition, $\underline{\underline{I}}_B$ is constant, then $\boldsymbol{M}_B = \underline{\underline{I}}_B \dot{\boldsymbol{\omega}}$, the three-dimensional counterpart of the planar law $M_B = I_B\dot{\omega}$; and if there is no torque, the angular momentum $\underline{\underline{I}}_B\boldsymbol{\omega}$ is conserved.

Why must the moment of inertia be a tensor, rather than a number? If the coordinate axes happen to be chosen such that $\underline{\underline{I}}_{\mathrm{cm}}$ is diagonal, the balance decouples into $M_{\mathrm{cm},i} = I_{\mathrm{cm},i}\dot{\omega}_i$, $i = 1, 2, 3$, and each axis has its own moment of inertia. In general, however, the angular momentum $\underline{\underline{I}}\boldsymbol{\omega}$ is not parallel to the angular velocity: the resistance of a body against rotation depends on the axis, and the linear map that captures this dependence is a second-order tensor.

Example 3.7 *(Moment of inertia of a cylinder).* Consider a homogeneous cylinder of radius R and length L, mass $M = \rho\pi R^2 L$, with its axis along x_3 and the origin at its centre of mass. In cylindrical coordinates $x_1 = r\cos\varphi$, $x_2 = r\sin\varphi$, x_3, with $\mathrm{d}V = r\,\mathrm{d}r\,\mathrm{d}\varphi\,\mathrm{d}x_3$,

$$[\underline{\underline{I}}_{\mathrm{cm}}] = \int_0^R \int_{-L/2}^{L/2} \int_0^{2\pi} \begin{bmatrix} r^2\sin^2\varphi + x_3^2 & -r^2\sin\varphi\cos\varphi & -rx_3\cos\varphi \\ & r^2\cos^2\varphi + x_3^2 & -rx_3\sin\varphi \\ \text{sym.} & & r^2 \end{bmatrix} \rho\, r\,\mathrm{d}\varphi\,\mathrm{d}x_3\,\mathrm{d}r.$$

All off-diagonal entries vanish, because $\int_0^{2\pi}\sin\varphi\cos\varphi\,\mathrm{d}\varphi = \int_0^{2\pi}\cos\varphi\,\mathrm{d}\varphi = 0$. For the axial moment, $I_{33} = \rho\cdot\frac{R^4}{4}\cdot 2\pi\cdot L = M\frac{R^2}{2}$. The transverse moment I_{11} consists of two parts: $\rho\frac{R^4}{4}\cdot\pi\cdot L = M\frac{R^2}{4}$ from the term $r^2\sin^2\varphi$, and $\rho\frac{R^2}{2}\cdot 2\pi\cdot\frac{L^3}{12} = M\frac{L^2}{12}$ from the term x_3^2. Hence

$$[\underline{\underline{I}}_{\mathrm{cm}}] = \begin{bmatrix} \frac{MR^2}{4} + \frac{ML^2}{12} & 0 & 0 \\ 0 & \frac{MR^2}{4} + \frac{ML^2}{12} & 0 \\ 0 & 0 & \frac{MR^2}{2} \end{bmatrix}.$$

◀

Two limiting cases of this result are used constantly in plane problems. A disk or cylinder rotating about its own axis has $I_{\mathrm{cm}} = \frac{1}{2}MR^2$, and a slender rod or bar ($R \to 0$) rotating about an axis through its centre perpendicular to it has $I_{\mathrm{cm}} = \frac{1}{12}ML^2$. In these planar applications we have tacitly assumed $\dot{I}_{\mathrm{cm}} = 0$. This is justified: for rotation about x_3 the balance reads $M_{\mathrm{cm},3} = \frac{\mathrm{d}}{\mathrm{d}t}(I_{\mathrm{cm},33}\omega_3)$, and $I_{\mathrm{cm},33} = \int_{\mathcal{B}}(\bar{x}_1^2 + \bar{x}_2^2)\rho\,\mathrm{d}V = \int_{\mathcal{B}}\bar{r}^2\rho\,\mathrm{d}V$ only involves the distances from the axis, which do not change in a plane rotation. Hence $M_{\mathrm{cm}} = I_{\mathrm{cm}}\dot{\omega}$.

Example *(Pulling up a weight).* A pulley of mass M and radius R on a fixed axle ($I_{\mathrm{cm}} = \frac{1}{2}MR^2$) is driven by a torque T and lifts a weight m by a rope. With the rope force S and the height $x = R\varphi$ of the weight,

$$m\ddot{x} = S - mg, \qquad I_{\mathrm{cm}}\ddot{\varphi} = T - SR.$$

Eliminating $S = T/R - \frac{1}{2}M\ddot{x}$ from the first equation gives

$$\ddot{x} = \frac{T/R - mg}{m + M/2}.$$

The pulley behaves like an additional mass $M/2$ that has to be accelerated together with the weight. ◀

3.2.3 Properties of the moment of inertia tensor

Computing moments of inertia by direct integration quickly becomes laborious. Fortunately the tensor has a number of properties that reduce most practical calculations to tabulated values and a little algebra.

Additivity. Since an integral over a union of regions is the sum of the integrals, the moment of inertia of a body composed of parts $\mathcal{B}_1$ and $\mathcal{B}_2$ is the sum of the moments of inertia of the parts – provided all of them refer to the *same* point B:

$$\underline{\underline{I}}_B = \underline{\underline{I}}_{B,1} + \underline{\underline{I}}_{B,2}. \tag{3.9}$$

Holes can be treated by subtraction. A hollow cylinder, for example, is a full cylinder of radius R_o minus one of radius R_i, and by Example 3.7

$$I_{11} = I_{22} = \frac{M_o}{12}(L^2 + 3R_o^2) - \frac{M_i}{12}(L^2 + 3R_i^2), \qquad I_{33} = \frac{M_o R_o^2}{2} - \frac{M_i R_i^2}{2},$$

with $M_{o,i} = \rho\pi R_{o,i}^2 L$. For a thin ring with $R_{o,i} = R \pm t/2$ and $t \to 0$ the axial moment becomes $I_{33} = \frac{\rho\pi L}{2}(R_o^2 + R_i^2)(R_o^2 - R_i^2) \to (\rho\, 2\pi R t L)\, R^2 = MR^2$, in agreement with Example 2.6.

Symmetry. For a homogeneous body, symmetry forces many entries of the tensor to vanish.

Proposition 3.8. *If the x_i-axis is an axis of rotational symmetry of the body, or if it is normal to a plane of symmetry, then all off-diagonal components with index i vanish (the reference point must lie on the axis or in the plane).*

Consider, for instance, a cone with its apex at B and its axis along x_1. Since x_1 is an axis of rotational symmetry, $I_{12} = I_{13} = 0$; since x_2 and x_3 are normal to planes of symmetry, also $I_{23} = 0$. The reason is easily seen in the integral $I_{12} = -\int_{\mathcal{B}} x_1 x_2 \rho \,\mathrm{d}V$: every slice $x_1 = \text{const}$ is a circle centred on the axis, and for each of its points with coordinate x_2 there is one with $-x_2$, so the contributions cancel. The same argument applies to $I_{23} = -\int x_2 x_3 \rho \,\mathrm{d}V$.

Positive definiteness and principal axes. For an arbitrary unit vector $\boldsymbol{n}$,

$$\boldsymbol{n} \cdot \underline{\underline{I}}_B \boldsymbol{n} = \int_{\mathcal{B}} \big[(\boldsymbol{r}\cdot\boldsymbol{r})(\boldsymbol{n}\cdot\boldsymbol{n}) - (\boldsymbol{n}\cdot\boldsymbol{r})(\boldsymbol{r}\cdot\boldsymbol{n})\big]\rho \,\mathrm{d}V = \int_{\mathcal{B}} \big[|\boldsymbol{r}|^2 - (\boldsymbol{r}\cdot\boldsymbol{n})^2\big]\rho \,\mathrm{d}V > 0,$$

because the projection of $\boldsymbol{r}$ onto $\boldsymbol{n}$ is never longer than $\boldsymbol{r}$ itself; in fact the integrand is the squared distance of the point from the axis through B in the direction $\boldsymbol{n}$. The tensor is therefore symmetric and positive definite, and linear algebra tells us that it can be diagonalised with real, positive eigenvalues.

Theorem 3.9 (Principal axes)**.** *For every body $\mathcal{B}$ and every point B there is a frame C in which*

$$[\underline{\underline{I}}_B]_C = \begin{bmatrix} I_1 & & \\ & I_2 & \\ & & I_3 \end{bmatrix}, \qquad I_1, I_2, I_3 > 0.$$

The axes of this frame, the eigenvectors of $\underline{\underline{I}}_B$, are the principal axes *of the body, and the I_k its principal moments of inertia. Axes of rotational symmetry and axes normal to symmetry planes are always principal axes.*

In practice one should always try to work in principal axes; with respect to the centre of mass the balance then takes the simple form $[\boldsymbol{M}_{\mathrm{cm}}] = (I_1\dot\omega_1,\, I_2\dot\omega_2,\, I_3\dot\omega_3)^{\mathsf T}$.

For plane motion only the component $I_{B,33} = \int_{\mathcal B}(x_1^2 + x_2^2)\rho\,\mathrm dV = \int_{\mathcal B} r^2\rho\,\mathrm dV$ is needed, and the balance reads $M_B = I_{B,33}\dot\omega$. For a long slender rod with its pivot B at one end, for example, x_2 is negligible across the thin cross-section A, and

$$I_B = \int_0^L x_1^2 \rho A\,\mathrm dx_1 = \frac{1}{3}\rho A L^3 = \frac{1}{3}ML^2.$$

The parallel axis theorem. Tables give moments of inertia with respect to the centre of mass. To transfer them to another point we use the following result, known in German as the *Satz von Steiner*.

Theorem 3.10 (Parallel axis theorem)**.** *Let* $\Delta\boldsymbol{x} = \boldsymbol{r}_B - \boldsymbol{r}_{\mathrm{cm}}$. *Then*

$$\underline{\underline{I}}_B = \underline{\underline{I}}_{\mathrm{cm}} + M\big[(\Delta\boldsymbol{x}\cdot\Delta\boldsymbol{x})\underline{\underline{\mathbb{1}}} - \Delta\boldsymbol{x}\,\Delta\boldsymbol{x}^{\mathsf T}\big], \tag{3.10}$$

that is, in components,

$$[\underline{\underline{I}}_B] = [\underline{\underline{I}}_{\mathrm{cm}}] + M\begin{bmatrix} \Delta x_2^2+\Delta x_3^2 & -\Delta x_1\Delta x_2 & -\Delta x_1\Delta x_3 \\ -\Delta x_1\Delta x_2 & \Delta x_1^2+\Delta x_3^2 & -\Delta x_2\Delta x_3 \\ -\Delta x_1\Delta x_3 & -\Delta x_2\Delta x_3 & \Delta x_1^2+\Delta x_2^2 \end{bmatrix}.$$

In plane motion, $I_B = I_{\mathrm{cm}} + M|\Delta\boldsymbol{x}|^2$.

Proof. Write $\boldsymbol{r} - \boldsymbol{r}_B = \boldsymbol{s} - \Delta\boldsymbol{x}$ with $\boldsymbol{s} = \boldsymbol{r} - \boldsymbol{r}_{\mathrm{cm}}$ and insert this into the definition of $\underline{\underline{I}}_B$. Expanding the products, the terms quadratic in $\boldsymbol{s}$ give $\underline{\underline{I}}_{\mathrm{cm}}$, the terms quadratic in $\Delta\boldsymbol{x}$ give the second term (with $\int \mathrm dm = M$), and the mixed terms are linear in $\boldsymbol{s}$ and vanish, since $\int_{\mathcal B}\boldsymbol{s}\,\mathrm dm = \boldsymbol{0}$ by the definition of the centre of mass. □

The proof shows that the theorem relates an arbitrary point to the *centre of mass*; to move between two arbitrary points one has to pass through the centre of mass. It also shows that, among all parallel axes, the one through the centre of mass has the smallest moment of inertia.

Two examples: for the slender rod, $I_{\mathrm{cm}} = \frac{1}{12}ML^2$ and $|\Delta\boldsymbol{x}| = L/2$ give $I_B = \frac{1}{12}ML^2 + \frac{1}{4}ML^2 = \frac{1}{3}ML^2$, in agreement with the direct integration above. For a wheel of radius R, $I_{\mathrm{cm}} = \frac{1}{2}MR^2$, and about its contact point A with the ground $I_A = I_{\mathrm{cm}} + MR^2 = \frac{3}{2}MR^2$. For a rolling wheel the angular momentum balance may be used in either of the forms $M_{\mathrm{cm}} = I_{\mathrm{cm}}\dot\omega$ and $M_A = I_A\dot\omega$ (see the remark after Example 3.10).

Relation to area moments of inertia. For a thin plate of thickness t lying in the x_1x_2-plane, the mass moments of inertia are proportional to the area moments of inertia known from beam theory. The polar moment is $I_p = \int r^2\,\mathrm dA = I_{33}/(\rho t)$; since x_3 is negligible across the plate, $I_{11} = \int(x_2^2 + x_3^2)\rho\,\mathrm dV \approx \rho t\int x_2^2\,\mathrm dA = \rho t\, I_x$ and similarly $I_{22} = \rho t\, I_y$; and the product of inertia is $I_{12} = -\rho t\, I_{xy}$.

The following example shows how the balance laws are combined in a system of several rigid bodies.

Example 3.10 *(Pulley system).* Pulley 1 (mass m, radius r, $I_{\mathrm{cm}} = \frac{1}{2}mr^2$) turns on a fixed axle at the ceiling and is driven by a force F acting tangentially on its rim. A

rope runs from pulley 1 down under the free pulley 2 (mass M, radius R) and back up to a fixed point on the ceiling. A weight of mass m_w hangs from the centre of pulley 2. We call the rope forces S_1 (between the pulleys), S_2 (to the ceiling) and S_3 (between pulley 2 and the weight).

The kinematics is governed by one observation: the rope segment leading to the ceiling does not move, so the point where it leaves pulley 2 is the ICR of that pulley. If pulley 2 and the weight move up by x, then $x = R\theta$, and the opposite side of pulley 2, at distance $2R$ from the ICR, moves twice as fast. The rope, and with it the rim of pulley 1, therefore moves by $2x$: $r\varphi = 2R\theta = 2x$. The system has one DOF.

For the weight, the momentum balance gives

$$m_w \ddot{x} = S_3 - m_w g. \tag{I}$$

For pulley 1, the angular momentum balance about its fixed centre, with $\ddot{\varphi} = 2\ddot{x}/r$, gives

$$\tfrac{1}{2} m r^2 \cdot \frac{2\ddot{x}}{r} = (F - S_1)\, r \qquad \Longrightarrow \qquad m\ddot{x} = F - S_1. \tag{II}$$

For pulley 2 we take moments about its ICR, which eliminates the unknown S_2. With $I_{\mathrm{ICR}} = \frac{3}{2} M R^2$ and $\ddot{\theta} = \ddot{x}/R$,

$$\tfrac{3}{2} M R^2\, \ddot{\theta} = S_1 \cdot 2R - (S_3 + Mg)\, R \qquad \Longrightarrow \qquad \tfrac{3}{2} M \ddot{x} = 2S_1 - Mg - S_3. \tag{III}$$

The combination $2 \cdot (\mathrm{II}) + (\mathrm{III}) + (\mathrm{I})$ eliminates S_1 and S_3:

$$\Big(2m + \tfrac{3}{2} M + m_w\Big)\ddot{x} = 2F - Mg - m_w g \qquad \Longrightarrow \qquad \ddot{x} = \frac{2F - (M + m_w)g}{2m + \frac{3}{2} M + m_w}.$$

◀

Remark 3.11. The angular momentum balance about a moving point in general contains additional terms, and the simple form $M = I\dot{\omega}$ about the ICR is not justified in general. For a rolling disk (or a pulley rolling on its rope) it is: the contact point is momentarily at rest and its acceleration points towards the centre of mass, and with these two properties the additional terms vanish. The form $M_A = I_A \dot{\omega}$ may therefore be used for rolling wheels, but it should not be applied blindly to other moving points.

3.2.4 The angular momentum transfer formula

The balance $\boldsymbol{M}_B = \dot{\boldsymbol{H}}_B + \boldsymbol{v}_B \times \boldsymbol{P}$ holds for any point B, which need not even belong to the body. It is often convenient to relate angular momenta about different points. Writing $\boldsymbol{r}_B = \boldsymbol{r}_{\mathrm{cm}} + \boldsymbol{r}_{\mathrm{cm},B}$,

$$\boldsymbol{H}_B = \int_{\mathcal{B}} (\boldsymbol{r} - \boldsymbol{r}_B) \times \boldsymbol{v}\, \mathrm{d}m = \int_{\mathcal{B}} (\boldsymbol{r} - \boldsymbol{r}_{\mathrm{cm}}) \times \boldsymbol{v}\, \mathrm{d}m - \boldsymbol{r}_{\mathrm{cm},B} \times \int_{\mathcal{B}} \boldsymbol{v}\, \mathrm{d}m = \boldsymbol{H}_{\mathrm{cm}} + \boldsymbol{P} \times \boldsymbol{r}_{\mathrm{cm},B}.$$

Writing the same relation for a second point A and subtracting, with $\boldsymbol{r}_{\mathrm{cm},B} - \boldsymbol{r}_{\mathrm{cm},A} = \boldsymbol{r}_{AB}$, we obtain:

Theorem 3.12 (Angular momentum transfer formula)**.**

$$\boldsymbol{H}_B = \boldsymbol{H}_A + \boldsymbol{P} \times \boldsymbol{r}_{AB}. \tag{3.11}$$

It has exactly the structure of the moment transfer formula of statics, $\boldsymbol{M}_B = \boldsymbol{M}_A + \boldsymbol{F} \times \boldsymbol{r}_{AB}$, with the force replaced by the momentum.

Example I *(Disk rolling without slipping).* A disk of mass M and radius R on rough ground (static friction coefficient μ_0) is pulled by a horizontal force F applied at its centre. The ground exerts the normal force N and a friction force F_R at the contact point A. The momentum balance in the horizontal direction is $Ma_{\mathrm{cm}} = F - F_R$. For the angular momentum balance we use the contact point, about which neither N nor F_R has a torque: $I_A\dot{\omega} = FR$ with $I_A = \frac{3}{2}MR^2$. Thus

$$\dot{\omega} = \frac{2}{3}\frac{F}{MR}, \qquad a_{\mathrm{cm}} = R\dot{\omega} = \frac{2}{3}\frac{F}{M}, \qquad F_R = F - Ma_{\mathrm{cm}} = \frac{1}{3}F.$$

Without friction the disk would slide without rotating, since F has no torque about the centre of mass; it is the friction force that makes the disk turn. Because the contact point is the ICR and does not slide, this is *static* friction, and rolling without slipping is only possible as long as $F_R \le \mu_0 N = \mu_0 Mg$, i.e. $F \le 3\mu_0 Mg$. ◀

Example II *(Spinning disk placed on the ground).* A disk spinning with $\omega(0) = \omega_0$ but with $v_{\mathrm{cm}}(0) = 0$ is placed on the ground (friction coefficient μ). At first it slips, so dynamic friction of magnitude $F_R = \mu Mg$ acts at the contact point, in the direction in which it pushes the centre forward. The momentum balance and the angular momentum balance about the centre of mass give

$$Ma_{\mathrm{cm}} = \mu Mg \;\Rightarrow\; v_{\mathrm{cm}}(t) = \mu g t, \qquad \tfrac{1}{2}MR^2\dot{\omega} = -\mu MgR \;\Rightarrow\; \omega(t) = \omega_0 - \frac{2\mu g}{R}\,t.$$

The disk speeds up while its rotation slows down, until the rolling condition $v_{\mathrm{cm}} = R\omega$ is reached at the time t_f; from then on the contact point no longer slides, friction becomes static (and in fact zero, since no horizontal force is needed to keep a disk rolling), and the disk rolls on uniformly. From $\mu g t_f = R\omega_0 - 2\mu g t_f$,

$$t_f = \frac{R\omega_0}{3\mu g}, \qquad v_{\mathrm{cm}}(t_f) = \frac{1}{3}R\omega_0, \qquad \omega(t_f) = \frac{1}{3}\omega_0.$$

There is an elegant way to obtain the final state without knowing the friction force. Take the angular momentum balance about the contact point A, which moves along with the disk. Its velocity is parallel to $\boldsymbol{P}$, so $\boldsymbol{v}_A \times \boldsymbol{P} = \boldsymbol{0}$; and all external forces either act at A or (gravity and N) cancel, so $\boldsymbol{M}_A = \boldsymbol{0}$. Hence $\boldsymbol{H}_A$ is conserved. (Note that $\boldsymbol{M}_A = \boldsymbol{0}$ does *not* imply $\dot{\omega} = 0$: while the disk slips the contact point is not the ICR, and $\boldsymbol{H}_A \neq I_A\boldsymbol{\omega}$.) By the transfer formula,

$$\begin{aligned}\boldsymbol{H}_A &= \boldsymbol{H}_{\mathrm{cm}} + \boldsymbol{P} \times \boldsymbol{r}_{\mathrm{cm},A} = -\tfrac{1}{2}MR^2\omega\,\boldsymbol{e}_3 + Mv_{\mathrm{cm}}\boldsymbol{e}_1 \times (-R\boldsymbol{e}_2)\\ &= -\big(\tfrac{1}{2}MR^2\omega + MRv_{\mathrm{cm}}\big)\boldsymbol{e}_3,\end{aligned}$$

so $\frac{R}{2}\omega + v_{\mathrm{cm}}$ is constant, and comparing the initial state with the rolling state $v_{\mathrm{cm}} = R\omega$ gives $\frac{R}{2}\omega_0 = \frac{3}{2}v_{\mathrm{cm}}$, i.e. $v_{\mathrm{cm}} = \frac{1}{3}R\omega_0$ as before. ◀

3.2.5 Work–energy balance

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To formulate the work–energy balance we first need the kinetic energy of a rigid body. Choose a reference point C of the body and insert the transfer formula

$\boldsymbol{v} = \boldsymbol{v}_C + \boldsymbol{\omega} \times (\boldsymbol{r} - \boldsymbol{r}_C)$ into $T = \frac{1}{2}\int_{\mathcal{B}} |\boldsymbol{v}|^2 \,\mathrm{d}m$. Expanding the square gives three terms:

$$T = \frac{1}{2}|\boldsymbol{v}_C|^2 \int_{\mathcal{B}} \mathrm{d}m + \boldsymbol{v}_C \cdot \Big(\boldsymbol{\omega} \times \int_{\mathcal{B}} (\boldsymbol{r} - \boldsymbol{r}_C)\,\mathrm{d}m\Big) + \frac{1}{2}\int_{\mathcal{B}} \left|\boldsymbol{\omega} \times (\boldsymbol{r} - \boldsymbol{r}_C)\right|^2 \mathrm{d}m.$$

The first integral is M. The second is $M(\boldsymbol{r}_{\mathrm{cm}} - \boldsymbol{r}_C)$, so the middle term vanishes if C is the centre of mass, and also if $\boldsymbol{v}_C = \boldsymbol{0}$, i.e. if C is the ICR (or lies on the IAR). For the last term we use $(\boldsymbol{\omega} \times \boldsymbol{a}) \cdot (\boldsymbol{\omega} \times \boldsymbol{a}) = \big[\boldsymbol{a} \times (\boldsymbol{\omega} \times \boldsymbol{a})\big] \cdot \boldsymbol{\omega}$, which by the computation of the previous section yields $\frac{1}{2}(\underline{\underline{I}}_C \boldsymbol{\omega}) \cdot \boldsymbol{\omega}$.

Theorem 3.13. *If C is the centre of mass or a point with $\boldsymbol{v}_C = \boldsymbol{0}$, the kinetic energy of a rigid body is*

$$T = \frac{1}{2} M |\boldsymbol{v}_C|^2 + \frac{1}{2}\,\boldsymbol{\omega} \cdot \underline{\underline{I}}_C \boldsymbol{\omega}, \tag{3.12}$$

the sum of a translational and a rotational part. In plane motion, $T = \frac{1}{2}Mv_C^2 + \frac{1}{2}I_C\omega^2$.

The work–energy balance of the system of particles then becomes

$$T(t_2) - T(t_1) = W_{12} = \sum_i \int_{\boldsymbol{r}_i(t_1)}^{\boldsymbol{r}_i(t_2)} \boldsymbol{F}_i^{\mathrm{ext}} \cdot \,\mathrm{d}\boldsymbol{r}_i + \int_{t_1}^{t_2} \boldsymbol{M}_T \cdot \boldsymbol{\omega}\,\mathrm{d}t, \tag{3.13}$$

where $\boldsymbol{r}_i$ is the point of application of the force $\boldsymbol{F}_i$, and external torques $\boldsymbol{M}_T$ contribute work as well, as we saw in Example 2.8. Internal forces do no work, since the body is rigid. As before, a conservative system satisfies $T + V = \text{const}$.

Example 3.11 *(Disk rolling up a hill).* A disk of mass M and radius R rolls without slipping with $v_0 = R\omega_0$ onto a slope. How high does it get?

The normal force and the friction force do no work: they act at the contact point, which is the ICR and therefore at rest. (Static friction on a stationary point does no work.) The system is thus conservative. The kinetic energy can be computed with respect to the centre of mass,

$$T = \frac{M}{2}v_{\mathrm{cm}}^2 + \frac{1}{2} \cdot \frac{MR^2}{2}\Big(\frac{v_{\mathrm{cm}}}{R}\Big)^2 = \frac{3}{4}Mv_{\mathrm{cm}}^2,$$

or, equivalently, with respect to the ICR, using Steiner's theorem: $T = \frac{1}{2}(\frac{1}{2}MR^2 + MR^2)\omega^2 = \frac{3}{4}Mv_{\mathrm{cm}}^2$. Energy conservation between the bottom and the highest point gives

$$\frac{3}{4}Mv_0^2 = Mgh_{\max} \qquad \Longrightarrow \qquad h_{\max} = \frac{3}{4}\,\frac{v_0^2}{g}.$$

A particle thrown upwards with the same speed would only reach $\frac{1}{2}v_0^2/g$; a considerable part of the kinetic energy of the disk is stored in its rotation. ◀

Example 3.12 *(Planetary gear).* A sun gear (radius r_s, moment of inertia $I_{\mathrm{cm},s}$) driven by a torque M_T meshes with three planet gears of radius r_p, which in turn mesh with a fixed outer ring. We seek the angular acceleration $\dot{\omega}_s$ of the sun gear; gravity plays no role because of the symmetry of the arrangement.

The contact forces between the gears act with equal magnitude and opposite direction on two points that move together, so they do no net work; the forces at the

fixed ring act at the ICRs of the planets and do no work either. The only work is done by the driving torque. The kinetic energy is

$$T = \frac{1}{2} I_{\mathrm{cm},s}\, \omega_s^2 + 3 \cdot \frac{1}{2} I_{\mathrm{ICR},p}\, \omega_p^2, \qquad I_{\mathrm{ICR},p} = I_{\mathrm{cm},p} + m_p r_p^2,$$

where we have used that each planet momentarily rotates about its contact point with the ring. The work–energy balance $T(t) - T(t_0) = \int M_T \,\mathrm{d}\varphi_s = M_T(\varphi_s - \varphi_{s,0})$, differentiated with respect to time, reads

$$I_{\mathrm{cm},s}\, \omega_s \dot{\omega}_s + 3 I_{\mathrm{ICR},p}\, \omega_p \dot{\omega}_p = M_T\, \omega_s.$$

It remains to relate ω_p to ω_s. The point of contact between sun and planet moves with the speed $\omega_s r_s$; for the planet this point is at the distance $2r_p$ from its ICR, so $\omega_s r_s = 2 r_p \omega_p$. Inserting $\omega_p = \omega_s r_s/(2r_p)$ and cancelling ω_s,

$$\dot{\omega}_s = \frac{M_T}{I_{\mathrm{cm},s} + 3 I_{\mathrm{ICR},p} \left(r_s/(2r_p)\right)^2}.$$

The planets act on the sun shaft like an additional, "reduced" moment of inertia. ◀

Example 3.13 *(A note on gears).* Gear 1 (radius r_1, moment of inertia I_1) is driven by a torque M_1 and drives gear 2 (radius r_2, moment of inertia I_2), which is loaded by a torque M_2; F_r is the force between the teeth. Since the gears roll on each other without slipping, $r_1\omega_1 = r_2\omega_2$. The angular momentum balances about the fixed axes are

$$I_1 \dot{\omega}_1 = M_1 - F_r r_1, \qquad I_2 \dot{\omega}_2 = F_r r_2 - M_2.$$

Eliminating F_r,

$$M_2 = M_1 \frac{r_2}{r_1} - \left(I_1 \dot{\omega}_1 \frac{r_2}{r_1} + I_2 \dot{\omega}_2 \right),$$

where the bracket represents the losses due to the inertia of the gears. For massless gears it vanishes, and $M_1/M_2 = r_1/r_2$, which together with the kinematic relation gives $M_1\omega_1 = M_2\omega_2$: the power is transmitted unchanged, and a gear simply trades torque for angular velocity. ◀